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Thesis Title: Effect of Duct-End Reflection Coefficient on
Axial-Fan Noise

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Effect of Duct-End Reflection Coefficient on Axial-Fan Noise

[ABSTRACT]: As the aviation industry has grown rapidly, aircraft noise has become an increasingly prominent concern. Fan noise from the engine, a core aircraft component, is a major source of aircraft noise. When designing or predicting the acoustic performance of duct-like systems such as engine inlets, the magnitude of the reflection coefficient at the duct opening provides an important measure of the system's sound radiation capability. Reflection also affects acoustic measurements at the duct wall. Investigating its effects on the in-duct sound field and far-field sound radiation can therefore inform engineering applications.

This thesis uses the finite element solver in the commercial software package COM-SOL to simulate ducts with different opening geometries. First, flanged ducts are modeled using the Pressure Acoustics interface. Reflection coefficients for different flange radii are obtained by acoustic impedance integration and compared with published results. The close agreement supports the feasibility of the simulation method. A bellmouth duct is then modeled using the Linearized Potential Flow interface, and the corresponding reflection coefficients are extracted by modal decomposition. Error perturbations are introduced into the simulated pressure data, and the resulting estimates are compared with existing experimental results to assess possible sources of discrepancy. The incident sound field obtained by modal decomposition is also reconstructed for verification. Finally, the influence of the reflection coefficient on standing waves inside the duct is investigated experimentally.

[Keywords]: Duct acoustics, Finite element method, Acoustic reflection coefficient, Modal decomposition

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1. Introduction

1.1 Research Background

On February 16, 1914, the world's first air route, between Los Angeles and San Francisco, opened. Commercial aviation has now developed for more than a century and plays an important role in the global economy. As the aviation industry has grown rapidly, aircraft noise has become an increasingly prominent issue. Over the past several decades, the number of airports has multiplied, making aircraft takeoff and landing noise unavoidable for people living nearby and sharply increasing public concern about aircraft noise.

To control aircraft noise, the US Congress first introduced the relevant regulation, FAR Part 36, in 1971, making civil aircraft noise a mandatory criterion for airworthiness certification. Since then, major aircraft manufacturers and aeronautical research institutions worldwide have devoted substantial efforts to aircraft noise research. Civil aircraft noise has continued to decrease as technology has advanced. Aircraft noise reduction is now a key technology in civil aviation, particularly for large passenger aircraft.

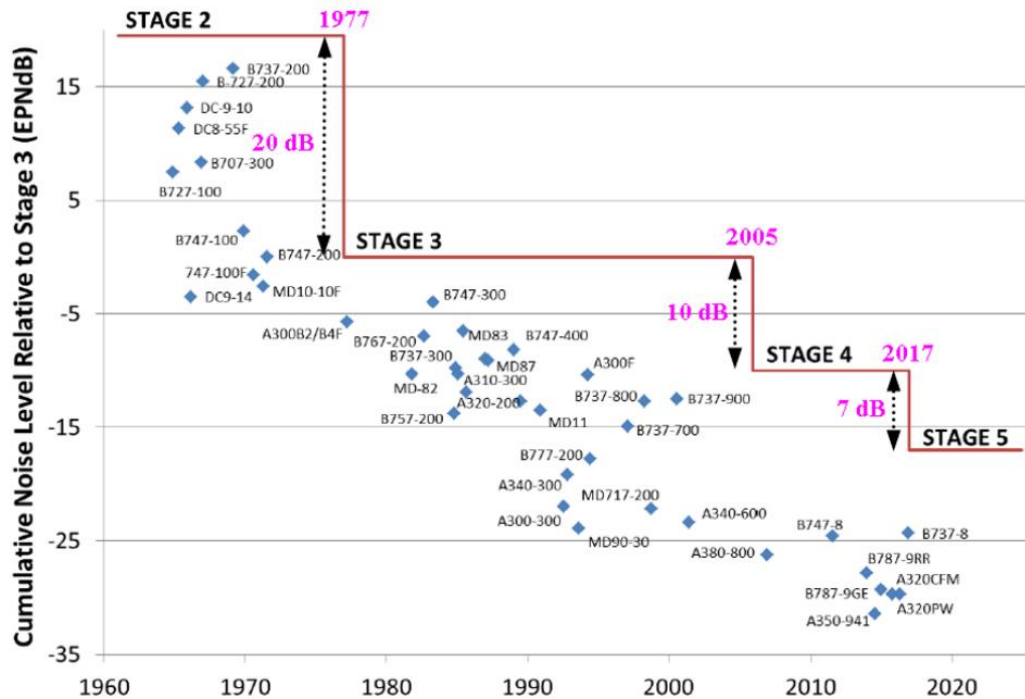


Figure 1.1. Trends in commercial aircraft noise [1]

According to their generation mechanisms, aircraft noise sources can be divided into

two broad categories: airframe noise and engine noise. The main noise sources of current aircraft are illustrated below.



Figure 1.2. Main noise sources of a civil aircraft

The contribution of each source to the total noise depends on the aircraft's operating condition. Using the Airbus A319 as a basis, Bertsch et al. evaluated the relative noise contributions for similar aircraft during takeoff and landing, as shown below. Under both conditions, the total engine noise sound pressure level is comparable to the overall aircraft noise sound pressure level. Reducing engine noise is therefore the principal task in reducing overall aircraft noise.

Current civil aeroengine noise can be divided into four categories: fan noise (rotor self-noise, rotor–stator interaction noise, shock noise, etc.), core noise (compressor and turbine noise), combustor noise, and jet noise (core and bypass jet noise, jet shock noise, etc.). During the early development of aeroengines, jet noise from turbojet engines was the dominant engine noise source. The subsequent introduction of turbofan engines changed this situation: they provided higher propulsive efficiency while substantially reducing jet velocity and consequently jet noise. Turbofans gradually replaced turbojets. As the bypass ratios of

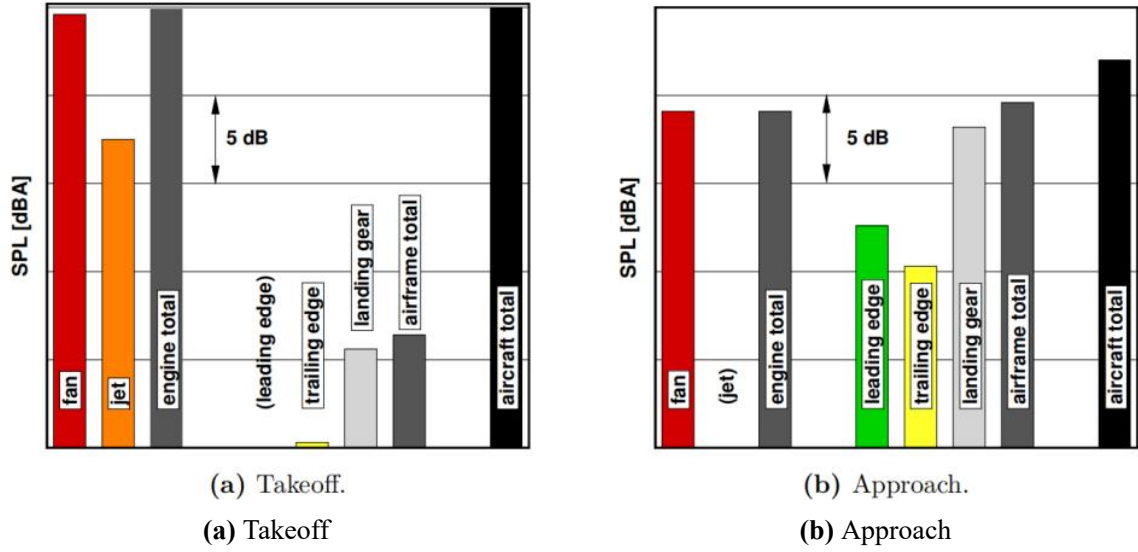


Figure 1.3. Contributions of the main noise sources of civil aircraft [2]

civil turbofan engines have continued to increase, fan dimensions and tangential velocities have also increased, and fan noise has gradually become the dominant engine noise source. Reducing fan noise is therefore the principal task in engine noise reduction.

1.2 Review of Previous Research

Engine fan noise measurement methods can be divided into two main categories: far-field measurements and duct-wall measurements.

1) Far-field measurements. This approach requires an anechoic chamber. A representative facility is the Aero-Acoustic Propulsion Laboratory (AAPL) at NASA Glenn Research Center. AAPL is a hemispherical anechoic chamber, with a cutoff frequency above 125 Hz and a 65-foot-high dome, used primarily for noise measurement experiments. In the 1990s, a height-adjustable, 4-foot-diameter Advanced Noise Control Fan (ANCF) rig [3] was established to design, test, and evaluate noise reduction technologies. From 1994 to 2016, ANCF operated in this enclosed anechoic environment, which supported far-field measurements. This enabled continuous acquisition of far-field data without moving the rig or otherwise affecting AAPL testing efficiency.

2) Duct-wall measurements. Noise measurements at the duct wall can clarify noise propagation within the duct, identify dominant sources, and investigate aerodynamic noise

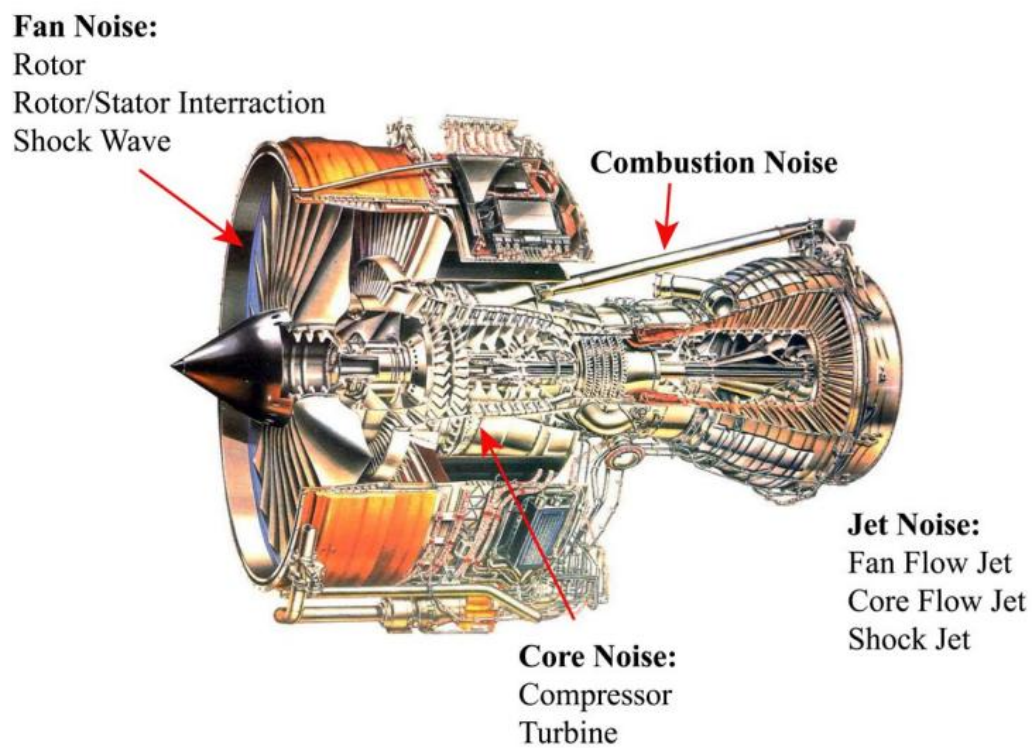


Figure 1.4. Main aeroengine noise sources



Figure 1.5. ANCF in AAPL [3]

generation mechanisms. The measured data can also be compared with simulations, providing a basis for engine noise reduction technologies. A representative example is the Source Diagnostic Test (SDT) conducted at NASA Glenn Research Center; acquiring broadband modal test data was one of its objectives. In the early 1980s, Boeing applied a wall-mounted microphone array to the inlet of a full-scale turbofan engine (JT15D) for modal measurements [4]. It later measured broadband fan noise using inlet and interstage arrays on an 18-foot fan rig [5]. Similar array arrangements were also used to measure blade-passing-frequency (BPF) noise from full-scale engines. Rademaker used a nonuniformly spaced array for modal measurements [6]. Premo and Joppa used a uniform microphone array in the inlet to decompose fan noise across modes and frequencies, obtaining the source diagnostic map below. The red dashed line schematically indicates the cut-off line. Strongly concentrated sources can be observed near this region; near the cutoff frequency, the reflection coefficient is also relatively large.

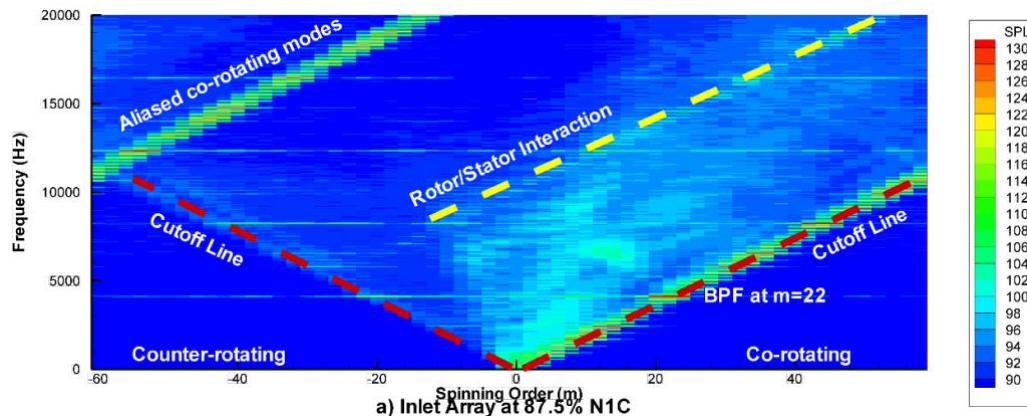


Figure 1.6. Aeroengine source diagnostic map [15]

The frequency dependence of the duct-opening reflection coefficient can be divided into three stages. Stage (1) is the cut-off stage, corresponding to the region below the red dashed line in Figure 1.6. No modes are cut on, and the reflection coefficient has no physical meaning. Stage (3) is the well cut-on stage, corresponding to the region far above the red dashed line in Figure 1.6, where the reflection coefficient has already become very small. Stage (2) is the cut-on stage, corresponding to the region around and above the red dashed

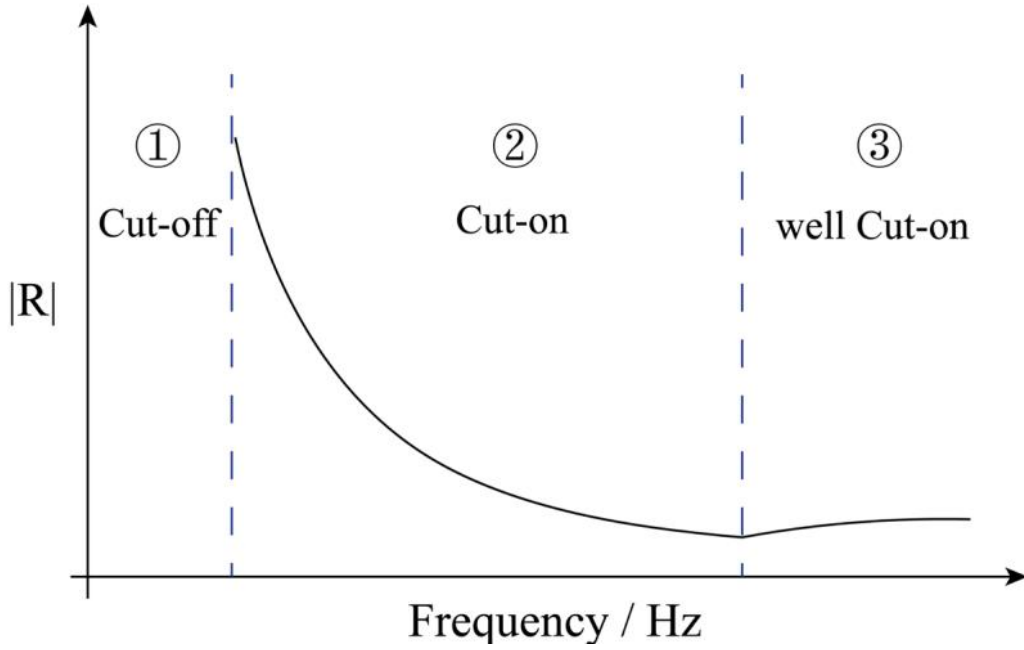


Figure 1.7. Schematic variation of the reflection coefficient

line in Figure 1.6. The part with relatively large reflection coefficients lies close to the red dashed line, which is precisely where sources are concentrated in an actual engine. Studying reflection coefficients in this stage is therefore particularly important.

Extensive research has addressed reflection coefficients in cylindrical ducts for commonly encountered acoustic modes. Levine and Schwinger obtained an exact solution for a thin-walled duct with an unflanged opening [7], while Nomura et al. derived an exact solution for a duct with an infinite flange [8]. Both are analytical solutions obtained through rigorous theoretical derivation. Using numerical methods, Caussé et al. [9] and Silva et al. [10] provided approximate formulas for unflanged duct reflection coefficients, and Norris and Sheng [11] provided an approximate expression for an infinitely flanged opening. A duct termination may also lie between these two limiting cases, having a flange of finite radius. Ando [12] first obtained an approximate expression for the reflection coefficient in this configuration. Based on numerical results, Dalmont et al. [13] proposed an end-correction formula that varies with the ratio of flange radius to duct radius. However, these results apply only at low frequencies. Silva et al. [14] investigated the variation of reflection coefficients with Helmholtz number for different ratios of flange radius to duct radius and provided

corresponding fitted curves.

As research has progressed, duct openings have evolved from simple flanges to curved shapes described by analytical expressions, such as aeroengine inlet bellmouths. Universal analytical solutions for the corresponding reflection coefficients have become difficult to obtain. Research has therefore shifted toward numerical solutions and converged asymptotic expressions, supported by experimental validation. Ville and Silcox [16] found experimentally that inlet shape (a straight opening or a bellmouth) affects acoustic radiation from the opening and the reflection coefficient near the cutoff frequency, whereas flow within the duct has little effect on radiation directivity. Most existing reflection-coefficient studies are limited to plane-wave or low-order circumferential-mode sources, with comparatively little work on high-order modes. In engineering measurements of sound pressure level at a particular axial position within a duct, however, reflection can introduce measurement errors, especially near the cutoff frequency where the reflection coefficient is large.

1.3 Scope of This Thesis

The main work of this thesis is to simulate ducts with different opening geometries using the Pressure Acoustics and Linearized Potential Flow interfaces in the Acoustics Module of the commercial COMSOL finite-element solver. Modal decomposition is then applied to the sound pressure data to obtain duct-opening reflection coefficients at different frequencies.

Chapter 2 introduces the basic theory of acoustic modes in cylindrical ducts, including the fundamentals of duct-mode decomposition, followed by the basic theory of reflection coefficients at circular duct openings. Three simple methods for calculating reflection coefficients are summarized, progressing from plane-wave to modal sources. Chapter 3 presents flanged-duct simulations using the COMSOL Pressure Acoustics interface. Reflection coefficients for different flange radii are compared with published data to validate the simulation approach. Chapter 4 uses the COMSOL Linearized Potential Flow interface to simulate a bellmouth duct. Error bands are introduced into the calculated reflection coefficients for comparison with existing experimental results. An extension investigates the reflection co-

efficient of the inlet of a high-bypass-ratio engine. Chapter 5 validates the sound field reconstruction results using the modal decomposition program. Chapter 6 presents standing-wave results inside the duct. Finally, Chapter 7 summarizes the work and outlines preliminary plans for future research.

2. Basic Theory

2.1 Theory of Cylindrical Acoustic Waveguides

In a homogeneous ideal fluid, the wave equation for sound pressure p in three-dimensional Cartesian coordinates is

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} = \frac{1}{c_0^2} \frac{\partial^2 p}{\partial t^2} \quad (2.1)$$

For compact notation, introducing the Laplacian operator ∇^2 gives the three-dimensional wave equation as

$$\nabla^2 p = \frac{1}{c_0^2} \frac{\partial^2 p}{\partial t^2} \quad (2.2)$$

For a cylindrical duct, the acoustic wave equation can be expressed in cylindrical coordinates:

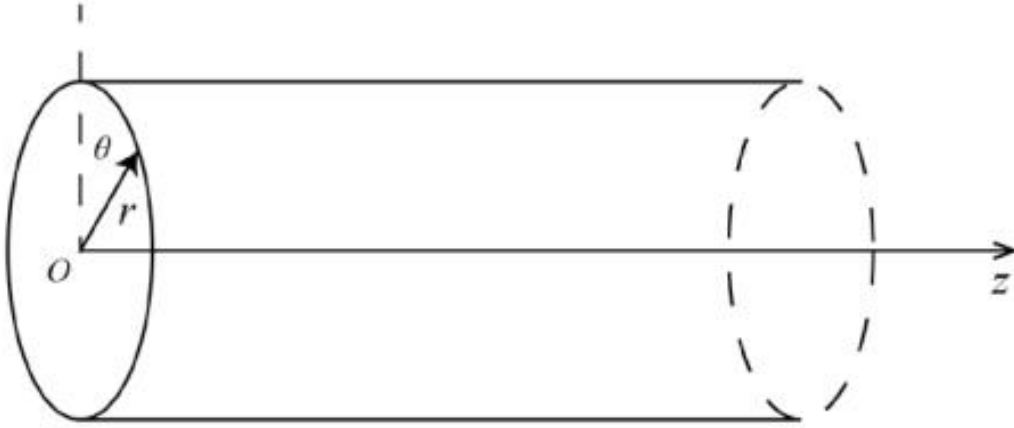


Figure 2.1. Cylindrical coordinate system used for the acoustic waveguide.

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ z &= z \end{aligned}$$

The Laplacian operator is then expressed as

$$\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} \quad (2.3)$$

The wave equation becomes

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 p}{\partial \theta^2} + \frac{\partial^2 p}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \quad (2.4)$$

The sound pressure solution inside the duct is

$$p_m(r, \theta, z) = A_m J_m(k_r r) \cos(m\theta - \varphi_m) e^{j(\omega t - k_z z)} \quad (2.5)$$

Here, $J_m(k_r r)$ denotes the cylindrical Bessel function of order m with argument $k_r r$, and k_r and k_z are the radial and axial wavenumbers, respectively.

For a rigid duct wall, the radial velocity at $r = a$ is zero:

$$v_{rm} = \frac{j}{\rho_0 \omega} \frac{\partial p_m}{\partial r} = 0 \quad (2.6)$$

That is,

$$\left[\frac{dJ_m(k_r r)}{d(k_r r)} \right]_{(r=a)} = 0$$

Using the recurrence relations for cylindrical Bessel functions,

$$\begin{aligned} \frac{dJ_m(x)}{dx} &= \frac{1}{2} [J_{m-1}(x) - J_{m+1}(x)] \\ \frac{dJ_0(x)}{dx} &= -J_1(x) \end{aligned} \quad (2.7)$$

we obtain

$$\begin{aligned} J_{m-1}(k_r a) &= J_{m+1}(k_r a) (m > 0) \\ J_1(k_r a) &= 0 (m = 0) \end{aligned} \quad (2.8)$$

These two equations yield a series of roots. For convenience, k_r is written as k_{mn} . Some roots calculated in MATLAB are listed below.

The sound pressure solution can therefore be written as

$$p_{mn}(r, \theta, z) = A_{mn} J_m(k_{mn} r) \cos(m\theta - \varphi_m) e^{j(\omega t - k_z z)} \quad (2.9)$$

Table 2.1. Values of $k_{mn}a$

$k_{mn}a$	m=0	m=1	m=2
n=0	0	1.841183781	3.054236928
n=1	3.83170597	5.331442774	6.706133194
n=2	7.01558667	8.536316366	9.969467823

The expression above describes a wave propagating in the positive z direction. For a wave propagating in the negative z direction, the final factor should instead be written as $e^{j(\omega t + k_z z)}$.

2.2 Cutoff Frequencies of Cylindrical Acoustic Waveguides

Consider a sound source of frequency f . If

$$\frac{2\pi f}{c_0} = k < k_{mn}$$

then

$$k_z = \pm \sqrt{k^2 - k_{mn}^2} = \pm j \sqrt{k_{mn}^2 - k^2}$$

$$p_{mn}(r, \theta, z) = A_{mn} J_m(k_{mn}r) \cos(m\theta - \varphi_m) e^{(j\omega t \mp \sqrt{k_{mn}^2 - k^2} z)}$$

In the limiting case $z \rightarrow +\infty$, $p_{mn}(r, \theta, +\infty)$ must remain finite. Accordingly, k_z in the expression above should be chosen as $-\sqrt{k^2 - k_{mn}^2} (= -j \sqrt{k_{mn}^2 - k^2})$, with a negative imaginary part. Thus,

$$p_{mn}(r, \theta, z) = A_{mn} J_m(k_{mn}r) \cos(m\theta - \varphi_m) e^{(j\omega t - \sqrt{k_{mn}^2 - k^2} z)} \quad (2.10)$$

When k_z is imaginary (more precisely, negative imaginary, $-j \sqrt{k_{mn}^2 - k^2}$), p_{mn} decays exponentially along the propagation direction. This is the cut-off state.

Conversely, if

$$\frac{2\pi f}{c_0} = k > k_{mn}$$

then

$$p_{mn} = A_{mn} \cos(m\theta - \varphi_m) J_m(k_{mn}r) e^{j(\omega t \mp \sqrt{k^2 - k_{mn}^2} z)} \quad (2.11)$$

In this case, p_{mn} can propagate without attenuation along the propagation direction. This is the cut-on state. The cutoff frequency of mode (m, n) is therefore

$$f_{cut-off} = \frac{k_{mn}c_0}{2\pi} = (k_{mn}a) \frac{c_0}{2\pi a} \quad (2.12)$$

The (m, n) acoustic mode can propagate only above this frequency; otherwise, only plane waves can propagate.

2.3 Acoustic Mode Propagation in a Cylindrical Duct with Flow

The preceding discussion concerns acoustic mode propagation in a duct without flow. This section considers a duct with flow. The analysis of acoustic mode propagation in a rigid-walled circular duct with flow adopts the following five assumptions:

- a. The flow is uniform and incompressible.
- b. Conditions are isentropic, and temperature gradients are neglected.
- c. The mean axial and circumferential velocity profiles remain unchanged, and the flow velocity is constant.
- d. There is no radial velocity distribution.
- e. The medium is uniformly distributed, with constant temperature and density.

Under these conditions, the solution of the acoustic wave equation in a rigid-walled cylindrical duct can be expressed as a linear superposition of acoustic modes [17]:

$$p(x, r, \theta) = \sum_{m=-\infty}^{\infty} \sum_{n=0}^{\infty} \left(P_{mn}^+ e^{-i\gamma_{mn}^+ x} + P_{mn}^- e^{-i\gamma_{mn}^- x} \right) \cdot J_m(k_{mn}r) \cdot e^{im\theta} \quad (2.13)$$

$$\gamma_{mn}^{\pm} = \frac{1}{1 - Ma^2} \left(\frac{Ma \cdot \omega}{c_0} \mp \kappa_{mn} \right) \quad (2.14)$$

$$\kappa_{mn}^2 + (1 - Ma^2)k_{mn}^2 = \left(\frac{\omega}{c_0} \right)^2 \quad (2.15)$$

Note that

$$\kappa_{mn} = \pm \sqrt{\left(\frac{\omega}{c_0} \right)^2 - (1 - Ma^2)k_{mn}^2} \quad (2.16)$$

Here, $\gamma_{mn}^{\pm}$ is the axial wavenumber, and the plus and minus signs indicate propagation direction: plus denotes downstream propagation and minus denotes upstream propagation. Assuming that the flow is in the positive x direction, $e^{-i\gamma_{mn}^+x}$ propagates downstream and the real part of γ_{mn}^+ is positive, whereas $e^{-i\gamma_{mn}^-x}$ propagates upstream and the real part of γ_{mn}^- is negative. Ma is the mean-flow Mach number; m and n denote the circumferential and radial mode orders, respectively. k_{mn} is the duct-mode eigenvalue, namely the root listed in Section 2.1 divided by the radius, $(k_{mn}a)/a$.

If κ_{mn} is real, the positive root in Equation (2.16) can be used directly. If κ_{mn} is imaginary, a negative imaginary part of $\gamma_{mn}^{\pm}$ gives a decaying wave, whereas a positive imaginary part gives an amplified wave. The sign of the imaginary part of $\gamma_{mn}^{\pm}$ can be controlled by choosing the sign of κ_{mn} , consistent with the discussion in Section 2.2.

Tyler and Sofrin [18] also noted that when the axial wavenumber $\gamma_{mn}^{\pm}$ is real, the (m, n) acoustic mode can propagate axially without attenuation and is cut on. When the axial wavenumber is complex, the mode decays exponentially along the axis and is cut off.

The critical frequency at which cut-on occurs can be calculated from

$$\left(\frac{\omega}{c_0} \right)^2 - (1 - Ma^2)\kappa_{mn}^2 = 0 \Rightarrow f_c = \frac{\kappa_{mn}}{2\pi} c_0 \sqrt{1 - Ma^2} \quad (2.17)$$

2.4 Acoustic Modal Decomposition in a Cylindrical Duct with Flow

For a set of sound pressure measurements obtained inside the duct, Equation (2.13) shows that the duct sound field is a linear superposition of circumferential and radial acoustic modes. Circumferential and radial modal decomposition are discussed separately below. Radial modal decomposition provides the incident sound pressure P_{mn}^+ and reflected sound pressure P_{mn}^- of mode (m, n) .

2.4.1 Circumferential Modal Decomposition

Considering only the circumferential acoustic modes in Equation (2.13) gives

$$p(x, r, \theta) = \sum_{m=-\infty}^{\infty} P_m(x, r) \cdot e^{im\theta} \quad (2.18)$$

The amplitude $P_m(x, r)$ of a circumferential mode at position (x, r) can be obtained by applying a discrete Fourier transform (DFT) to sound pressure data $p(x, r, \theta_c)$ sampled at different circumferential angles at that position [19]:

$$P_m(x, r) = \frac{1}{C} \sum_{c=1}^C p(x, r, \theta_c) e^{-im\theta_c} \quad (2.19)$$

According to the Nyquist sampling theorem [20], the number of circumferential measurement points C and the circumferential mode order M to be resolved must satisfy $C > 2M$.

2.4.2 Radial Modal Decomposition

A given circumferential mode can be regarded as a linear superposition of radial modes. Radial modal decomposition is performed for a specified circumferential mode. The complex amplitude of that circumferential mode at position (x, r) can be expressed as

$$P_m(x, r) = \sum_{n=0}^{\infty} \left(P_{mn}^+ e^{-i\gamma_{mn}^+ x} + P_{mn}^- e^{-i\gamma_{mn}^- x} \right) \cdot J_m(k_{mn}r) \quad (2.20)$$

For the finite number of radial modes that are cut on in the duct, the complex amplitude of the specified circumferential mode can be expressed as

$$P_m(x, r) = \sum_{n=0}^{N-1} \left(\mathbf{P}_{mn}^+ e^{-i\gamma_{mn}^+ x} + \mathbf{P}_{mn}^- e^{-i\gamma_{mn}^- x} \right) \cdot J_m(k_{mn}r) \quad (2.21)$$

That is,

$$\begin{aligned} P_m &= A^\pm P_{mn}^\pm \\ A^\pm &= \begin{bmatrix} e^{-i\gamma_{m0}^\pm x} J_m(k_{m0}r) & \dots & e^{-i\gamma_{mN-1}^\pm x} J_m(k_{mN-1}r) \\ P_{mn}^\pm = [P_{m0}^\pm & \dots & P_{mN-1}^\pm]^T \end{bmatrix}^T \end{aligned}$$

When forward- and backward-propagating waves are decomposed separately, the matrices above are written as

$$\mathbf{A} = [\mathbf{A}^+ \quad \mathbf{A}^-] = \begin{bmatrix} e^{-i\gamma_{m0}^+ x} J_m(k_{m0}r) & e^{-i\gamma_{m0}^- x} J_m(k_{m0}r) \\ \vdots & \vdots \\ e^{-i\gamma_{mN-1}^+ x} J_m(k_{mN-1}r) & e^{-i\gamma_{mN-1}^- x} J_m(k_{mN-1}r) \end{bmatrix} \quad (2.22)$$

$$\mathbf{P}_{mn} = [\mathbf{P}_{mn}^+ \quad \mathbf{P}_{mn}^-]^T = \begin{bmatrix} P_{m0}^+ & P_{m0}^- \\ \vdots & \vdots \\ P_{mN-1}^+ & P_{mN-1}^- \end{bmatrix}^T \quad (2.23)$$

Then,

$$P_m(x, r) = \mathbf{A} \mathbf{P}_{mn} \quad (2.24)$$

$\mathbf{P}_{mn}$ can be solved using the MATLAB pseudoinverse function, ‘pinv’.

The decomposition position specified above is (x, r) . For radial modal decomposition, two measurement-point arrangements are available:

(1) Measurement points at different radial positions

This arrangement disturbs the aerodynamics, so measurement points are generally placed on the wall wherever possible in experiments. In simulations, data can be extracted at different radial positions for analysis. From the standpoint of solving the equation system,

the number of radial measurement points D and the number of radial modes N to be resolved must satisfy $D \geq N$.

(2) Measurement points at different axial positions

In this arrangement, the radial position is fixed, usually at the wall, $r = a$. The required number of axial measurement positions Z and the number of radial modes N to be resolved generally satisfy [21] $Z \geq 2N$. To separate incident and reflected waves, the requirement is $Q \geq 4N$.

In addition, accurate duct-mode decomposition requires the spacing between axial measurement points on the wall to satisfy [22]

$$0.1\pi < \frac{ks}{1 - M^2} < 0.8\pi \quad (2.25)$$

Here, k is the axial wavenumber, s is the axial microphone spacing, and M is the Mach number. This equation will be used in Section 6.2 to assess the sound field reconstruction results.

2.5 Basic Theory of Duct-Opening Reflection Coefficients

Consider a plane wave propagating through a duct with cross-sectional area A and length l . At its open termination, the radiation impedance causes part of the acoustic wave to be reflected and part to be transmitted. A silencer is a typical example of this situation.

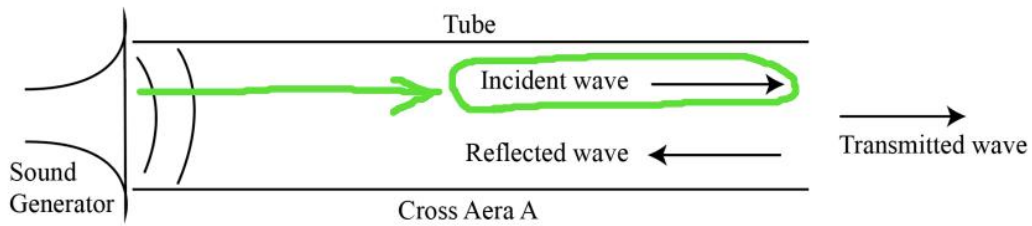


Figure 2.2. Schematic of acoustic propagation inside a duct

The incident and reflected sound pressures inside the duct are

$$p_i = P_i e^{j(\omega t - kx)} = P_i e^{j\omega t} e^{-jkx} = P_i(\omega) e^{-jkx} \quad (2.26)$$

$$p_r = P_r e^{j(\omega t + kx)} = P_r e^{j\omega t} e^{jkx} = P_r(\omega) e^{jkx} \quad (2.27)$$

The sound pressure reflection coefficient at the duct opening ($x = l$) is

$$R(\omega) = \frac{P_r(\omega)}{P_i(\omega)} e^{2jkl} = \frac{P_r}{P_i} e^{2jkl} = |R| e^{j\frac{2\omega l}{c}} \quad (2.28)$$

Here, $2\omega l/c$ is the phase difference between the incident and reflected waves at the interface. Three methods for calculating the reflection coefficient are introduced below.

2.5.1 Acoustic Impedance Integration Method

The incident and reflected plane waves are

$$p_i(x, t) = P_i e^{i\omega(t - \frac{x}{c})} \quad (2.29)$$

$$p_r(x, t) = P_r e^{i\omega(t + \frac{x}{c})} \quad (2.30)$$

Here, P_i and P_r are the incident and reflected sound pressure amplitudes, respectively.

At the duct-opening cross section ($x = l$), the sound pressure is

$$p(l, t) = P_i e^{i\omega(t - \frac{l}{c})} + P_r e^{i\omega(t + \frac{l}{c})} \quad (2.31)$$

The velocity continuity condition is

$$u(l, t) = \frac{1}{\rho_0 c} \left[P_i e^{i\omega(t - \frac{l}{c})} - P_r e^{i\omega(t + \frac{l}{c})} \right] \quad (2.32)$$

The acoustic impedance condition is

$$P_i e^{i\omega(t - \frac{l}{c})} + P_r e^{i\omega(t + \frac{l}{c})} = Z \cdot \frac{1}{\rho_0 c} \left[P_i e^{i\omega(t - \frac{l}{c})} - P_r e^{i\omega(t + \frac{l}{c})} \right] \quad (2.33)$$

Therefore,

$$R(\omega) = \frac{P_r}{P_i} = \left(\frac{\frac{Z}{\rho_0 c} - 1}{\frac{Z}{\rho_0 c} + 1} \right) e^{-\frac{i2\omega l}{c}} \quad (2.34)$$

Here, Z is the radiation impedance, defined as the ratio of the boundary-averaged pressure p to the boundary-averaged normal velocity v :

$$Z = \frac{p}{v} = \frac{\frac{1}{\pi a^2} 2\pi \int_0^a r p dr}{\frac{1}{\pi a^2} 2\pi \int_0^a r v dr} = \frac{\int_0^a r p dr}{\int_0^a r v dr} \quad (2.35)$$

It can be calculated in COMSOL by selecting the corresponding integration boundary.

Silva et al. further transformed the preceding equation (3.1.6) to obtain [10]

$$R = \frac{i \tan \left[\arctan \left(\frac{Z_{\Delta}(f)}{i Z_0} \right) - k \Delta \right] - 1}{i \tan \left[\arctan \left(\frac{Z_{\Delta}(f)}{i Z_0} \right) - k \Delta \right] + 1} \quad (2.36)$$

This expression is used to calculate the reflection coefficient at the duct outlet. Here, Δ is the distance between the impedance-extraction cross section and the duct opening, and Z_0 is the characteristic acoustic impedance inside the duct, $\rho_0 c / A$.

2.5.2 Two-Point Transfer-Function Method

The transfer-function method is generally used in impedance tubes to calculate the absorption coefficient α of sound-absorbing materials, as illustrated below.

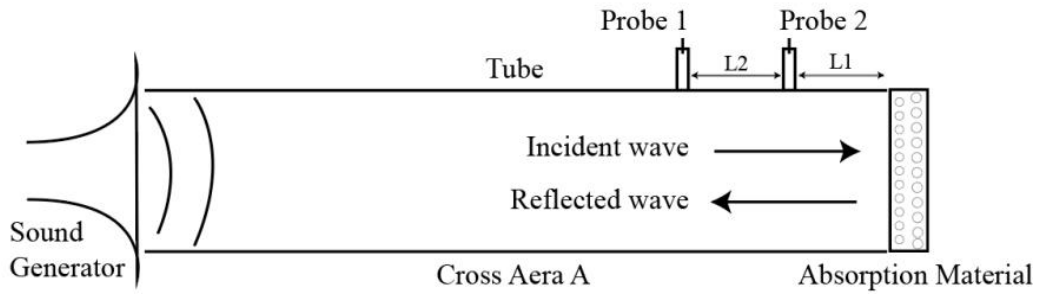


Figure 2.3. Measurement of the absorption coefficient of a sound-absorbing material using the two-point transfer-function method

Select measurement points 1 and 2 inside the duct at distances $L_1 + L_2$ and L_1 from the opening, respectively. Decompose the plane-wave sound field into an incident wave $\dot{p}_I$ and a reflected wave $\dot{p}_R$. The sound pressures at points 1 and 2, p_1 and p_2 , respectively, are

$$p_1 = P_I e^{jk(L_1+L_2)} + P_R e^{-jk(L_1+L_2)} \quad (2.37)$$

$$p_2 = P_I e^{jkL_1} + P_R e^{-jkL_1} \quad (2.38)$$

The transfer functions of the incident and reflected waves, H_I and H_R , respectively, are

$$H_I = \frac{p_{2I}}{p_{1I}} = e^{-jkL_2} \quad (2.39)$$

$$H_R = \frac{p_{2R}}{p_{1R}} = e^{jkL_2} \quad (2.40)$$

The complex transfer function from point 1 to point 2, H_{12} , is

$$H_{12} = \frac{p_2}{p_1} = \frac{e^{jkL_1} + r e^{-jkL_1}}{e^{jk(L_1+L_2)} + r e^{-jk(L_1+L_2)}} \quad (2.41)$$

The reflection coefficient r is written as

$$R = \frac{H_{12} - H_I}{H_R - H_{12}} e^{jk(L_1+L_2)} \quad (2.42)$$

2.5.3 Acoustic Modal Decomposition Method

The two reflection-coefficient extraction methods introduced above apply only to plane-wave sources and present certain difficulties in experiments:

1) For Method 1, the derivation assumes a plane wave, whereas the experimental source is generally a $(1, 0)$ mode. Moreover, cross-sectional integration cannot be performed in the experiment to obtain acoustic impedance data.

2) Method 2 is intended to measure the absorption coefficients of acoustic materials using a standard impedance tube. Outside that configuration, calculations based on only two measurement points have limited reliability.

This section therefore introduces a more reliable reflection-coefficient extraction

method applicable to different modal sources.

Section 2.4 discussed the theory of acoustic mode propagation and decomposition in circular ducts. The sound pressure inside a duct can be regarded as a superposition of sources with different circumferential and radial modes:

$$p(x, r, \theta) = \sum_{m=-\infty}^{\infty} \sum_{n=0}^{\infty} \left(\mathbf{P}_{mn}^{+} e^{-i\gamma_{mn}^{+}x} + \mathbf{P}_{mn}^{-} e^{-i\gamma_{mn}^{-}x} \right) \cdot J_m(k_{mn}r) \cdot e^{im\theta} \quad (2.43)$$

For the finite number of radial modes that are cut on in the duct, the complex amplitude of a specified circumferential mode can be expressed as

$$P_m(x, r) = \sum_{n=0}^{N-1} \left(\mathbf{P}_{mn}^{+} e^{-i\gamma_{mn}^{+}x} + \mathbf{P}_{mn}^{-} e^{-i\gamma_{mn}^{-}x} \right) \cdot J_m(k_{mn}r) \quad (2.44)$$

That is,

$$P_m = A^{\pm} P_{mn}^{\pm}$$

For a given circumferential mode, radial modal decomposition yields the incident and outgoing modal sound pressures:

$$\begin{aligned} \mathbf{P}_{mn}^{+} &= [P_{m0}^{+} \quad \cdots \quad P_{mN-1}^{+}] \\ \mathbf{P}_{mn}^{-} &= [P_{m0}^{-} \quad \cdots \quad P_{mN-1}^{-}] \end{aligned}$$

The modal reflection coefficient is then

$$R_{mn} = \frac{P_{mn}^{-}}{P_{mn}^{+}} \quad (2.45)$$

For a given acoustic mode, the reflection coefficient generally decreases as frequency increases. For convenience, frequency can be nondimensionalized as the Helmholtz number:

$$ka = \frac{\omega}{c}a = \frac{2\pi f}{c}a \quad (2.46)$$

The reflection coefficient then depends only on two dimensionless quantities: the mode

order and the Helmholtz number.

2.6 Summary

This chapter began with the theory of acoustic mode propagation in a cylindrical duct without flow and introduced the cutoff frequencies of cylindrical acoustic waveguides. It then presented acoustic mode propagation in a rigid-walled circular duct with flow under five assumptions. On this basis, methods for circumferential and radial modal decomposition in a cylindrical duct with flow were introduced. Finally, three methods for calculating reflection coefficients were described: impedance integration, suitable for simulation postprocessing; the two-point transfer-function method, used to measure absorption coefficients of acoustic materials; and acoustic modal decomposition, a more reliable method applicable to different modal sources. These three methods will be used in the following chapters to calculate reflection coefficients for different duct-opening geometries.

3. Simulation Study of Flanged Ducts

The main objective of this chapter is to model and simulate flanged ducts using the COMSOL Pressure Acoustics, Frequency Domain interface and reproduce published results to validate the simulation approach.

3.1 Governing Equation of the Pressure Acoustics Interface

The Pressure Acoustics interface describes the sound field using pressure p alone. It is mainly used to simulate acoustic problems in fluids, such as water and air, in which waves are exclusively or predominantly longitudinal. Viscous and thermal losses are neglected, and there is no flow. The frequency-domain governing equation, the Helmholtz equation, is

$$\nabla \cdot \left(-\frac{1}{\rho_0} \nabla p \right) - \frac{1}{\rho_0} \left(\frac{\omega}{c} \right)^2 p = 0 \quad (3.1)$$

3.2 Simulation Setup

3.2.1 Basic Parameters

Table 3.1. Parameters of the acoustic medium (air)

Density ρ_0	Sound speed c_0
1.25kg/m ³	343m/s

Table 3.2. Geometric parameters

Duct length L	Radius a	Flange radius b
0.5m	0.04m	0.2m

3.2.2 Modeling and Meshing

Simulating the exterior sound field requires a perfectly matched layer (PML), which allows an infinite region to be represented by a finite domain. The PML permits an open boundary to be simulated after spatial truncation, enabling the exterior sound field to be solved. The model is shown below.

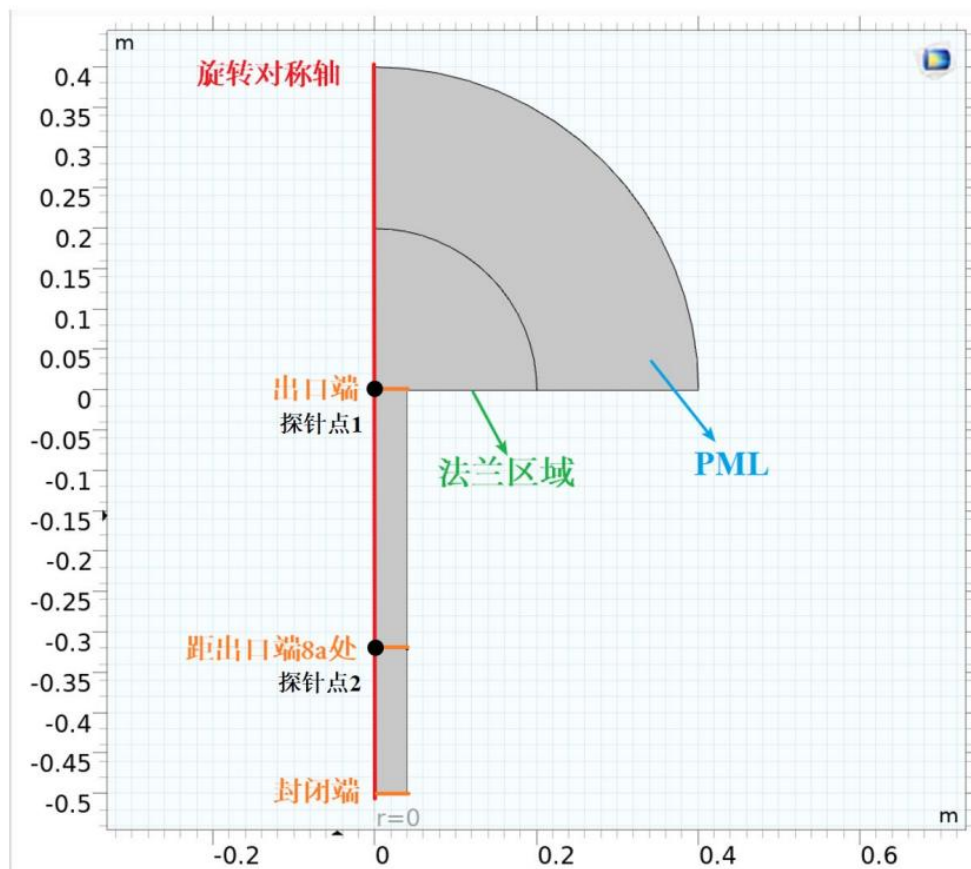


Image labels: axis of rotational symmetry (red); duct opening (orange); measurement point 1 (black); measurement point 2 (black) at a distance of $8a$ from the opening; closed end (orange); flange region (orange); PML (blue).

Figure 3.1. Flanged-duct model

For the infinite-flange case discussed in reference [14], the computational domain can be constructed using the simulation setup in reference [23], which studied reflection coefficients at flanged duct openings with different angles, as shown below.

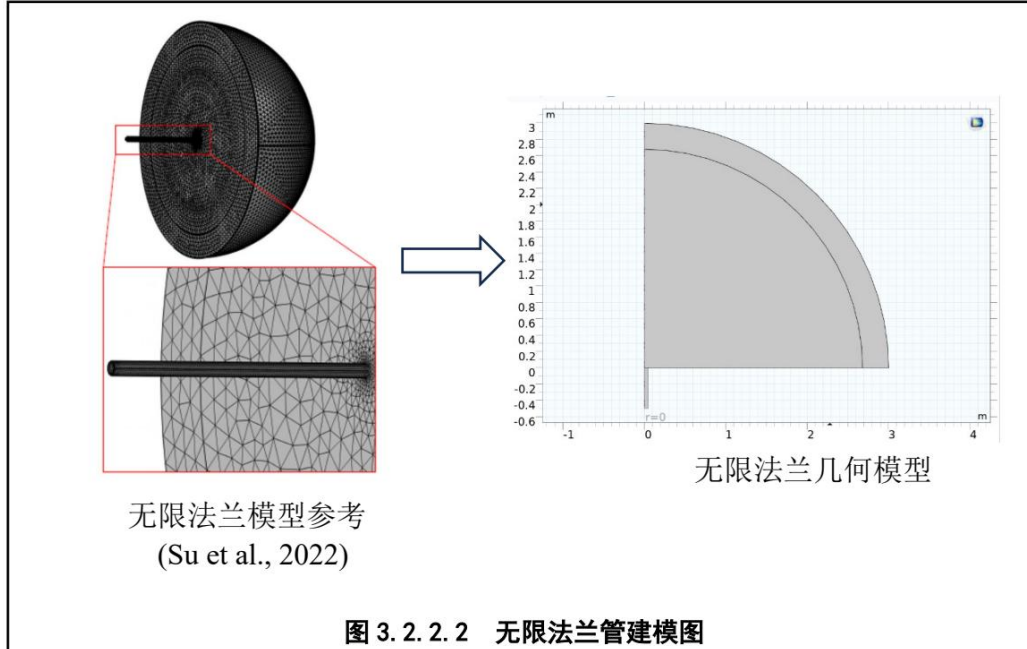


Image labels: reference model with an infinite flange (Su et al., 2022); infinite-flange geometric model.

Figure 3.2. Computational model of a duct with an infinite flange [23].

Twenty mesh elements are used per shortest wavelength. A free triangular mesh is used in all regions except the PML.

The modeling approach above was found to produce substantial deviations from published data when the flange radius was smaller than the duct radius. A possible reason is that the PML is too close to the opening in this configuration: the PML should be placed at a distance of two to four times the characteristic structural dimension or greater. Consequently, acoustic waves may not be adequately absorbed.

To avoid this issue, the computational domain is extended behind the flange to simulate unflanged ducts and flanged ducts with different radii, as shown below. The meshing approach remains unchanged.

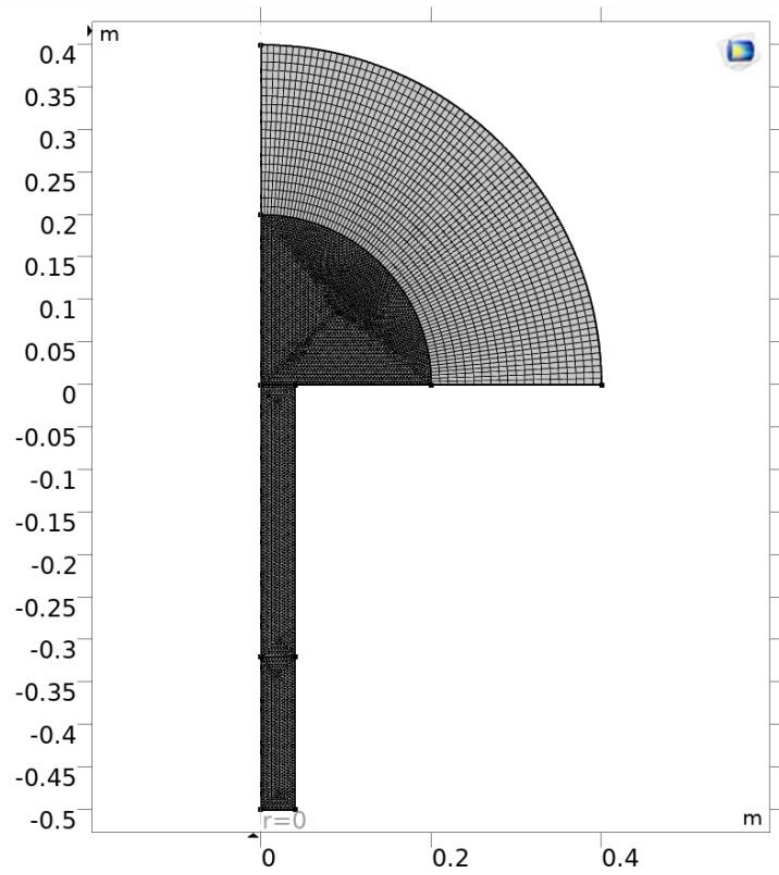


Figure 3.3. Flanged-duct mesh

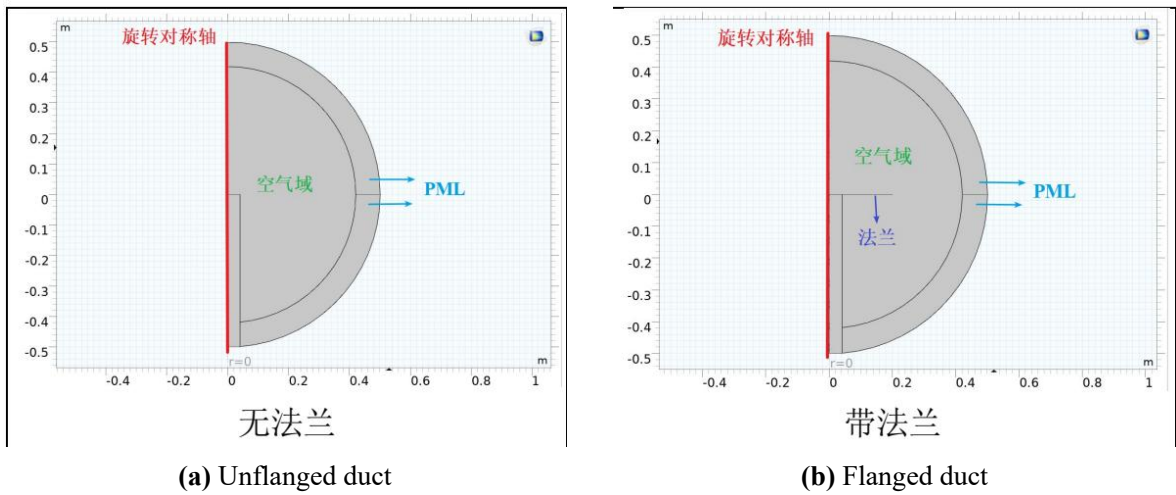


Image labels: axis of rotational symmetry (red); air domain (green); flange (purple); PML (blue); unflanged and flanged configurations.

Figure 3.4. Improved flanged-duct model

3.2.3 Physics Conditions

The setup follows the open-pipe example in the COMSOL Application Library:

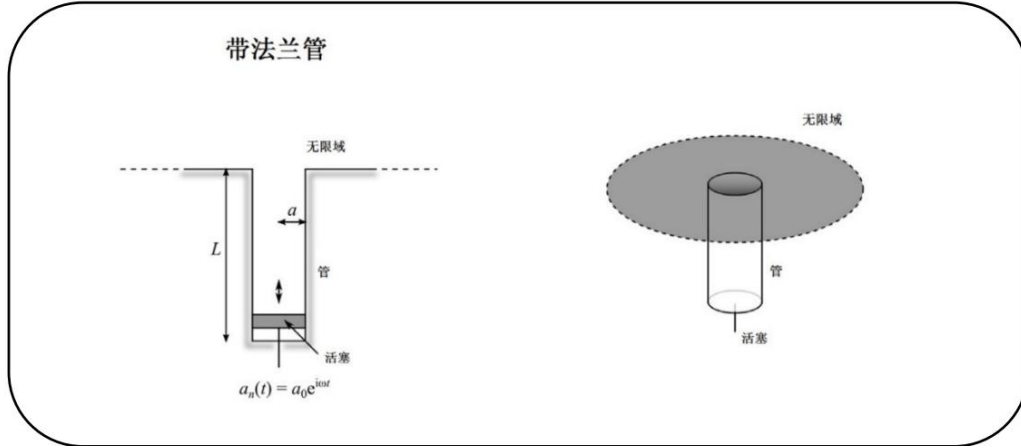


Image labels: flanged duct; infinite domain; duct; piston.

Figure 3.5. Schematic of the COMSOL open-pipe example

For axisymmetric source vibration, the cutoff frequency is $f_c = f_{01} = k_{01}c_0/(2\pi) = 3.832c_0/(2\pi a)$, where $ka = 3.832$. The closed end is modeled as a piston vibrating axially with inward velocity $V_n = 1$ m/s.

Because this is a theoretical model, the incident mode is set to $m = 0$, corresponding to a plane wave. Here, the cutoff frequency is determined using the first radial eigenvalue for $m = 0$. In the experiment, however, the eigenvalue for $m = 1$ is generally used to determine cutoff, giving $ka = 1.84$.

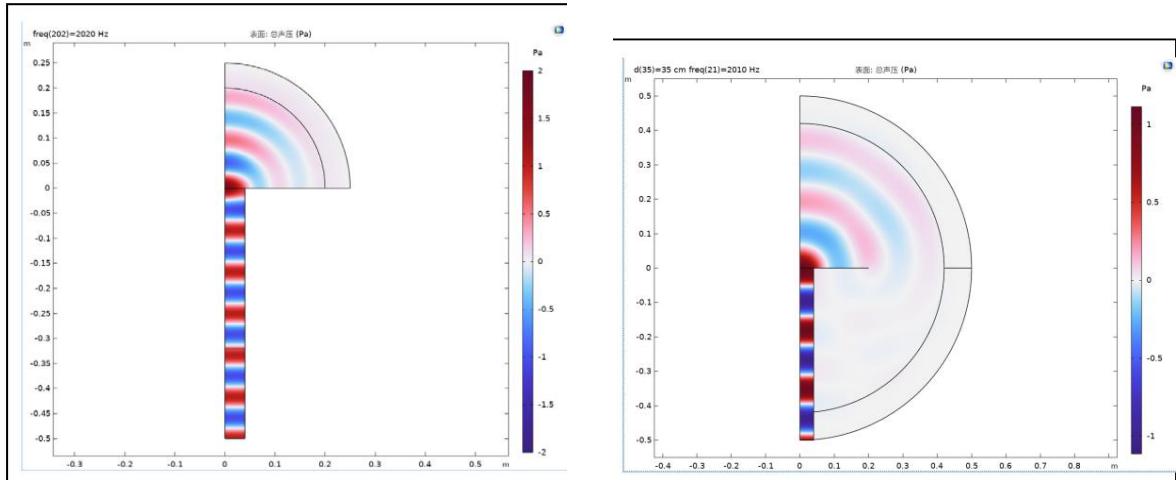
3.3 Reflection-Coefficient Results

The modeling approaches above yield the corresponding sound fields. Results from two modeling approaches are selected and plotted below.

3.3.1 Acoustic Impedance Integration Method

Reflection coefficients calculated using Equation (2.36) are compared with published results below.

Reflection-coefficient magnitude versus ka for unflanged and infinitely flanged ducts



(a) Initial computational domain (b) Extended computational domain
COMSOL plot labels: surface pressure (Pa).

Figure 3.6. Sound fields of flanged ducts

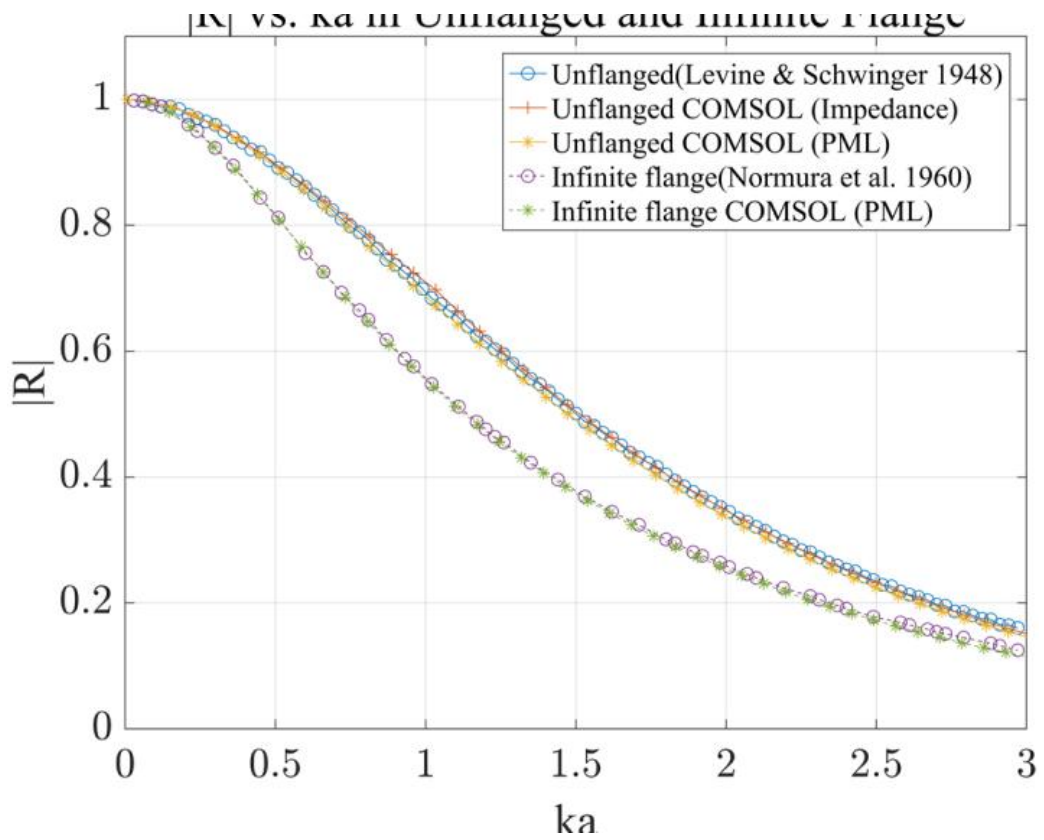
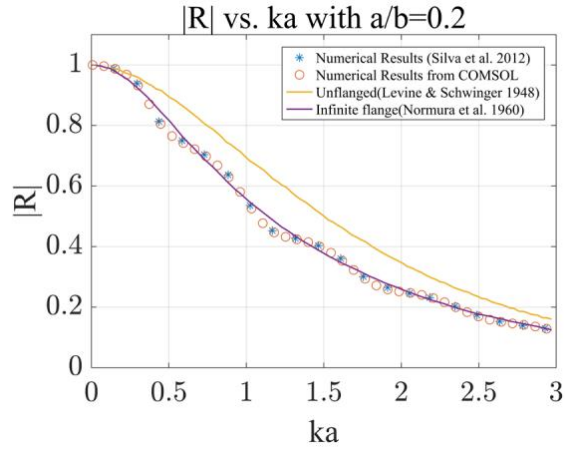
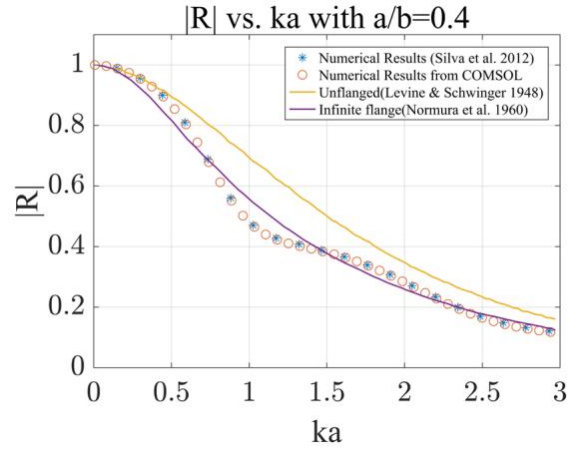


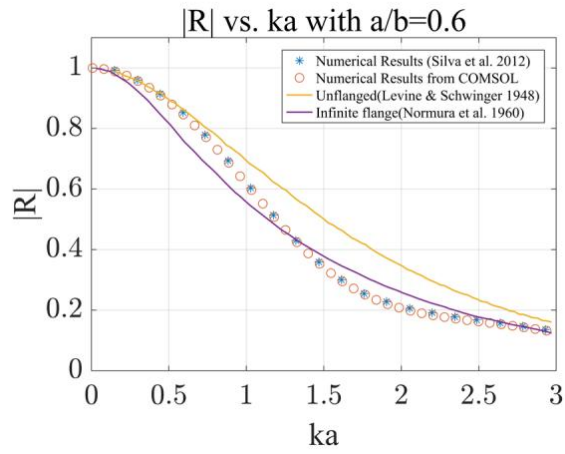
Figure 3.7. Comparison of simulated and published reflection coefficients for infinitely flanged and unflanged ducts [7][8]



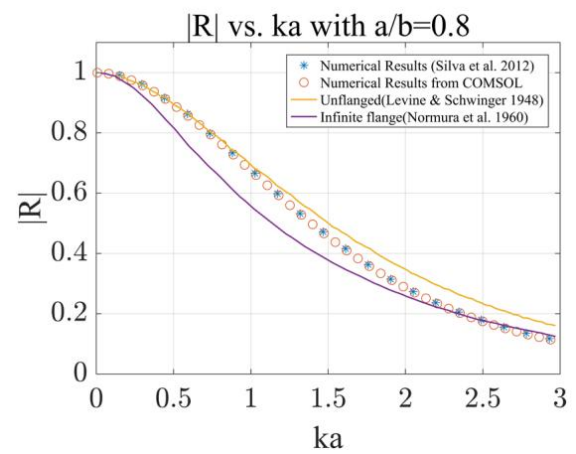
(a) $a/b = 0.2$



(b) $a/b = 0.4$



(c) $a/b = 0.6$



(d) $a/b = 0.8$

Figure 3.8. Comparison of simulation and published data for different flange radii [14]

As shown above, the data obtained using these modeling approaches agree closely with the published data, supporting the feasibility of the simulation method.

The reflection coefficients in this section are extracted from impedance obtained by integrating over a cross section. Such cross-sectional integration cannot be performed in an actual experiment. Measurements are generally taken sufficiently far from the opening that reflected evanescent modes have largely decayed. In actual experiments, the selected measurement position can also affect the results.

3.3.2 Two-Point Transfer-Function Method

Reflection coefficients of flanged ducts can also be calculated using the measurement-point method. With a point spacing of 3 cm, reflection coefficients were calculated at different positions for a flanged duct with $a/b = 0.2$. The results are shown below.

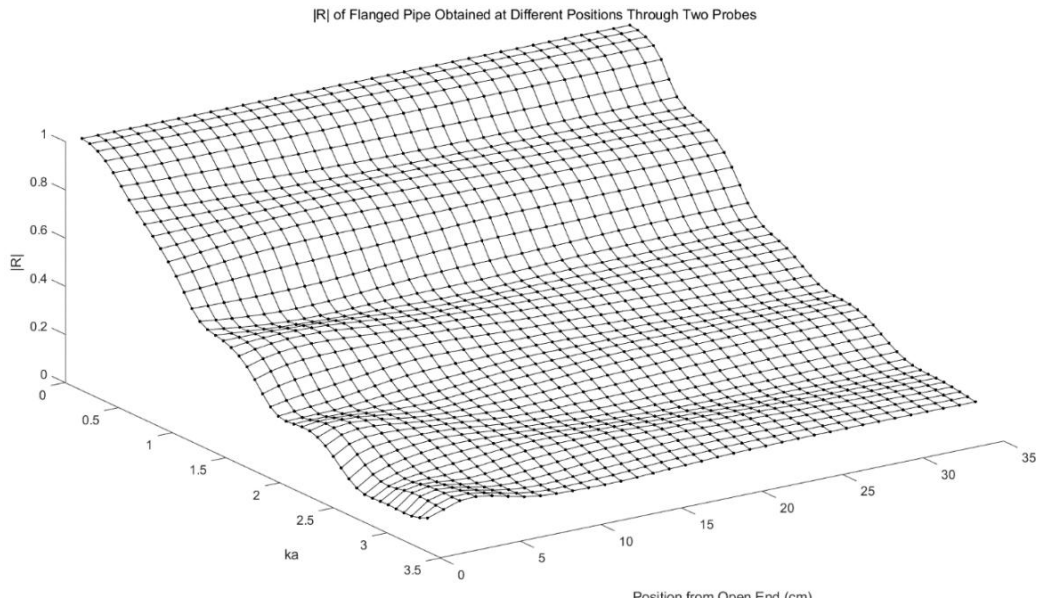


Figure 3.9. Reflection coefficients calculated from measurement points for the flanged duct with $a/b = 0.2$

The two-point transfer-function approach can be used experimentally in an impedance tube to calculate a specimen's absorption coefficient. However, when this method was applied to a bellmouth duct with an imposed $(1, 0)$ incident mode, the calculated reflection coefficient exceeded the experimental result by 0.2, which was outside the acceptable range. Be-

cause finite-element simulations provide full-field data, using only two measurement points does not make effective use of the available data. The following chapter therefore extracts reflection coefficients using modal decomposition and employs the Linearized Potential Flow, Frequency Domain interface with background flow.

3.4 Summary

This chapter modeled and simulated flanged ducts using the COMSOL Pressure Acoustics, Frequency Domain interface. Reflection coefficients calculated from the simulated sound pressures using acoustic impedance integration agreed very well with published results. The two-point transfer-function method was also used to calculate reflection-coefficient curves at different positions, but the results differed substantially from the experimental data. The following chapter will therefore use modal decomposition to extract reflection coefficients and the Linearized Potential Flow, Frequency Domain interface to account for background flow.

4. Simulation Study of a Bellmouth Duct

The main objective of this chapter is to model and simulate a bellmouth duct using the COMSOL Linearized Potential Flow, Frequency Domain interface. Some experimental reflection-coefficient results are already available for this duct, allowing comparison and analysis of the simulation data obtained here.

4.1 Governing Equations of the Linearized Potential Flow Interface

The COMSOL aeroacoustics interfaces address acoustic problems with background flow, namely the effects of flow on the sound field. An absorptive muffler is a classic example in which mean flow significantly affects acoustic performance. Linearized Potential Flow is a simplified interface particularly suited to describing compressible, fully isentropic, inviscid, and irrotational flow.

The Euler equations for an ideal gas undergoing an adiabatic process can be expressed as

$$\frac{\partial \tilde{\rho}}{\partial t} + \nabla \cdot (\tilde{\rho} \tilde{\mathbf{v}}) = 0 \quad (4.1)$$

$$\tilde{\rho} \left(\frac{\partial \tilde{\mathbf{v}}}{\partial t} + \tilde{\mathbf{v}} \cdot \nabla \cdot \tilde{\mathbf{v}} \right) + \nabla \tilde{p} = 0 \quad (4.2)$$

The equation of state and sound speed are

$$\gamma \tilde{p} = \tilde{\rho}^\gamma \quad (4.3)$$

$$\tilde{c}^2 = \frac{d\tilde{p}}{d\tilde{\rho}} = \tilde{\rho}^{\gamma-1} \quad (4.4)$$

For irrotational flow,

$$\tilde{\mathbf{v}} = \nabla \tilde{\phi} \quad (4.5)$$

The fundamental space- and time-dependent variables describing the flow are then the velocity potential $\tilde{\phi}$ and density $\tilde{\rho}$.

Introducing perturbations φ , $\mathbf{v}$, and ρ gives

$$\tilde{\phi} = \phi + \varphi e^{i\omega t} \quad (4.6)$$

$$\tilde{\mathbf{v}} = \mathbf{v} + \mathbf{v} e^{i\omega t} \quad (4.7)$$

$$\tilde{\rho} = \rho_0 + \rho e^{i\omega t} \quad (4.8)$$

When these perturbations are very small relative to their respective mean quantities, the equations can be linearized, giving the following linearized potential-flow equations:

$$i\omega\rho + \nabla \cdot (-\rho_0 \nabla \phi + \rho \mathbf{v}) = 0 \quad (4.9)$$

$$\rho_0(i\omega\phi + \mathbf{v} \cdot \nabla \phi) = p \quad (4.10)$$

$$p = c_0^2 \rho \quad (4.11)$$

4.2 Simulation Setup

4.2.1 Basic Parameters

Table 4.1. Ambient parameters

Background pressure P_0	Background temperature T_0	Background density ρ_0	Sound speed c_0
101325Pa	299.15K	1.1799kg/m ³	346.73m/s

In the experiment, the fan rotational speed rpm and corresponding flow velocity U_0 obey the following fitted relationship:

Table 4.2. Geometric parameters

Outer diameter r	Wall thickness	PML boundary length L1	Lined-section length L
0.1m	0.01m/3	0.4m	0.15m
Rigid-wall length upstream of the liner L2	Rigid-wall length downstream of the liner L3	Straight rigid-wall length downstream of the liner L4	
0.526m	0.8226m	0.7500m	

$$U_0 = 0.00327rpm - 0.1853 \quad (4.12)$$

$$rpm = \frac{60f}{B}$$

Here, f is the frequency of interest in Hz, and B is the number of blades, equal to 15 in the experiment. The equations above provide the flow-velocity input for simulations at different frequencies.

The inlet sound pressure level SPL_{in} was also measured at different rotational speeds. After conversion to a sound pressure amplitude, it can be prescribed as the incident sound pressure P_{in} in the simulation. However, the incident amplitude does not affect extraction of the reflection coefficient. The subsequent simulations therefore prescribe $P_{in} = 11.347108$ Pa.

4.2.2 Modeling and Meshing

The two-dimensional axisymmetric model is shown below.

Twelve mesh elements are used per shortest wavelength of interest. A structured mesh is used inside the duct, a mapped mesh in the PML, and a free triangular mesh elsewhere. The mesh is shown below.

The following points should be noted:

(1) For an elastic medium, finite-element method (FEM) simulations can be highly accurate [24].

(2) Structured meshes are preferred for regular geometries, such as the duct interior

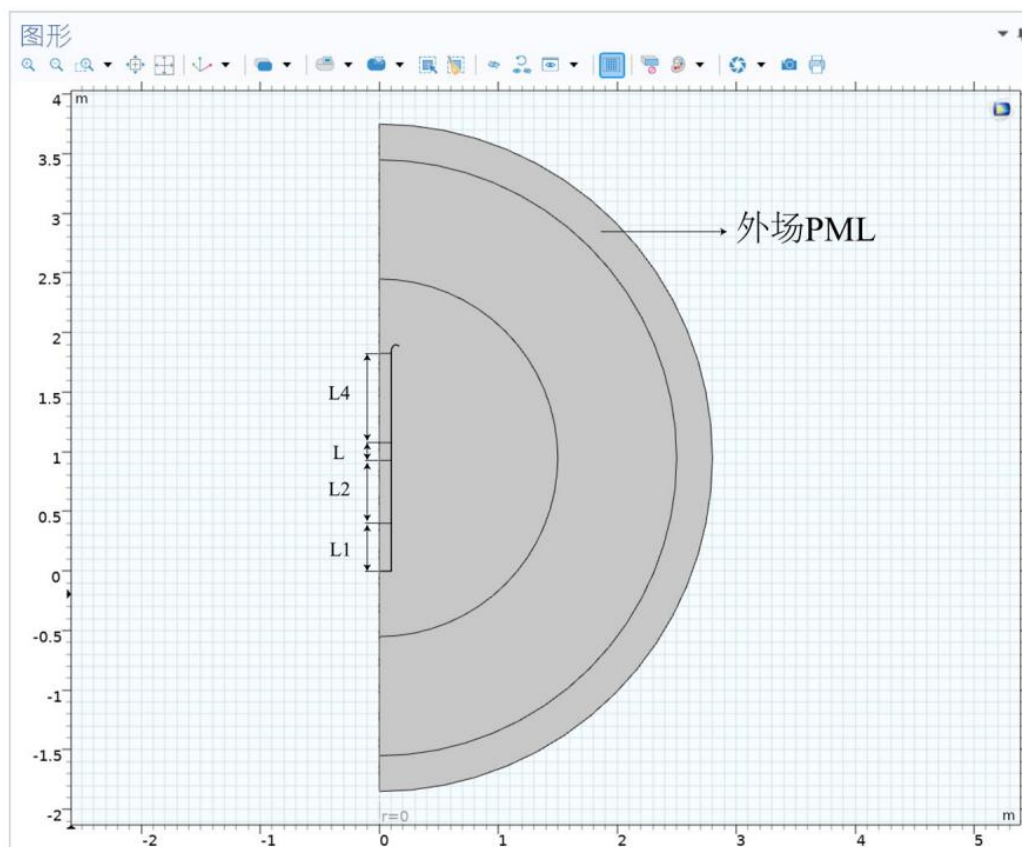


Image label: exterior PML.

Figure 4.1. Geometric model

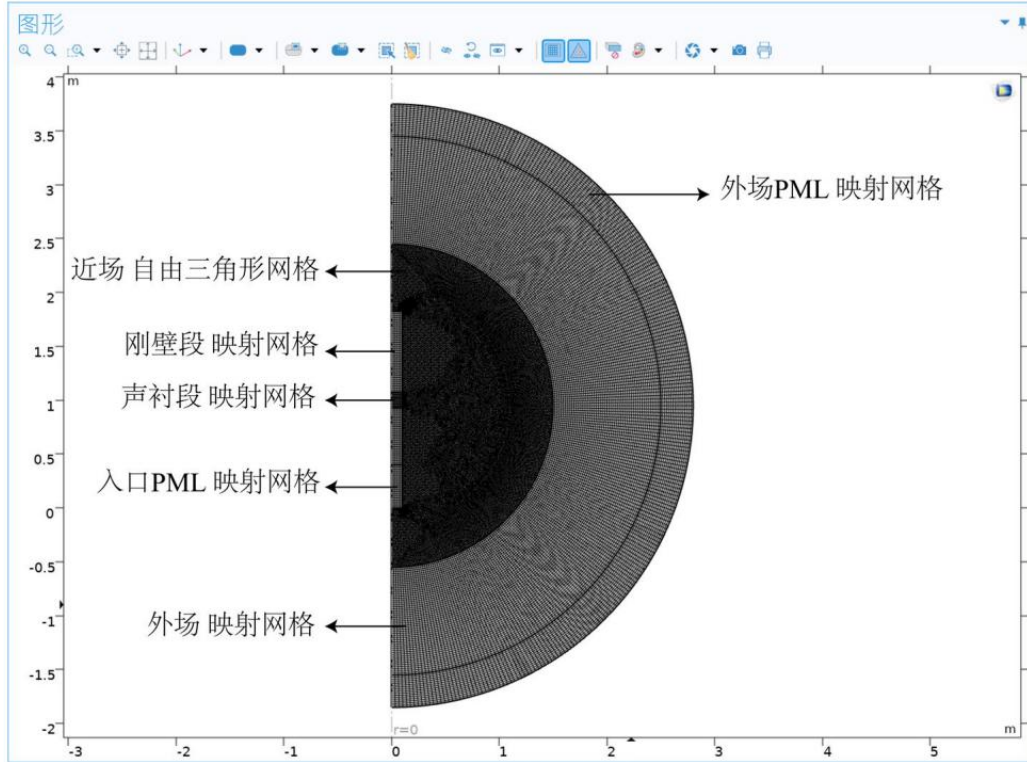


Image labels: exterior PML mapped mesh; near-field free triangular mesh; rigid-wall-section mapped mesh; liner-section mapped mesh; inlet PML mapped mesh; exterior-field mapped mesh.

Figure 4.2. Computational mesh

shown above.

(3) In acoustic simulations, at least 10 mesh elements are generally required to resolve the shortest wavelength.

(4) The PML absorbs acoustic waves. Its absorption performance is independent of its physical depth but depends on the number of layers along the acoustic propagation direction, generally at least eight. A mapped mesh must be used, or a swept mesh in three dimensions.

4.2.3 Physics Conditions

At the duct inlet, an incident sound pressure condition must be prescribed on the source surface. Experiments have verified that only one radial mode is present in this bellmouth duct, so only one (m, n) mode needs to be prescribed in the simulation. Equation (2.13) gives the incident source-surface pressure $p_{in}(m)$ for a specified circumferential mode. In a two-dimensional axisymmetric model, the circumferential-angle factor can be omitted, giving

$$p_{in}(m) = P_{in} \cdot J_m(k_{mn}r) \quad (4.13)$$

Using

$$\begin{aligned} \frac{\partial}{\partial t} &\rightarrow i\omega \\ \frac{\partial}{\partial z} &\rightarrow -ik_z \end{aligned}$$

the sound field in a duct with mean-flow velocity U_0 can be described by the convected Helmholtz equation [25]:

$$\frac{1}{c^2} \left(i\omega + U \frac{\partial}{\partial x} \right)^2 \phi - \nabla^2 \phi = 0 \quad (4.14)$$

The relationship below follows from the momentum equation.

The velocity potential and sound pressure are related by

$$p = -\rho_0 \left(i\omega + U \frac{\partial}{\partial x} \right) \phi \quad (4.15)$$

Thus,

$$\phi = \frac{p}{-i\rho_0(\omega - Uk_z)}$$

The inlet conditions are specified as follows.

Table 4.3. Parameters of the inlet conditions

Source-surface sound pressure $p_{in}(m)$ [Pa]	Velocity potential ϕ [m^2/s]	Axial velocity v_z [m/s]	Inward mass flux $m_{flow-in}$ [$kg/m^2/s$]
$P_{in} \cdot J_m(k_{mn}r)$	$\frac{p}{-i\rho_0(\omega - Uk_z)}$	$-ik_z\phi$	$\rho_0 v_z$

The corresponding physics setup is shown below.

4.3 Reflection-Coefficient Results

The modeling approach above yields sound fields for different incident acoustic modes. Four incident modes are selected here to illustrate the sound fields; these test cases use an

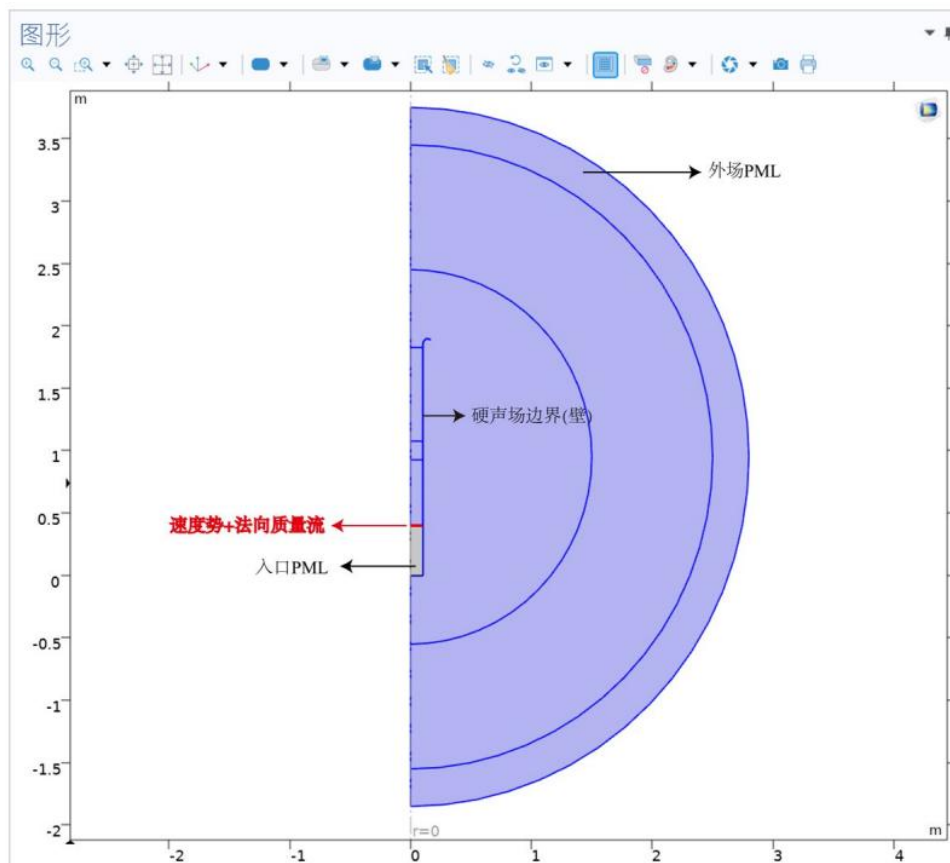
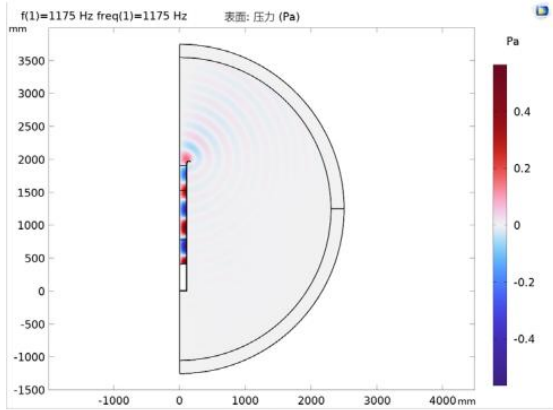


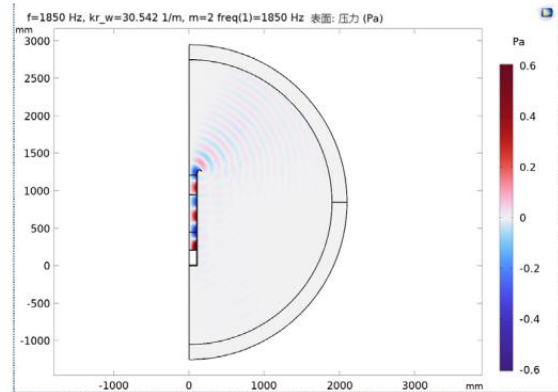
Image labels: exterior PML; hard boundary (wall); sound-pressure source applied on the boundary; inlet PML.

Figure 4.3. Physics setup

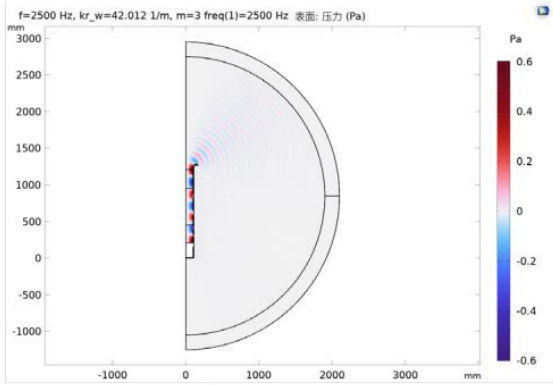
incident sound pressure amplitude of 1 Pa.



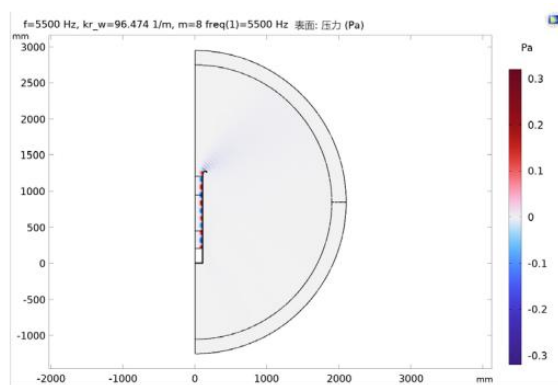
(a) (1, 0) mode



(b) (2, 0) mode



(c) (3, 0) mode



(d) (8, 0) mode

COMSOL plot label: surface pressure (Pa).

Figure 4.4. Bellmouth sound pressure fields for different modes

(1, 0) modal source

(2, 0) modal source

(3, 0) modal source

(8, 0) modal source

The corresponding fan test rig is shown below. Measurement-point positions and the associated experimental results will be introduced later.

4.3.1 Radial Decomposition

Sound pressure data are extracted along three cut lines at different radial positions inside the duct, $r = 0.02, 0.05$, and 0.08 m, extending from $L1 + L2 + L + 0.01$ to $L1 + L2 + L +$

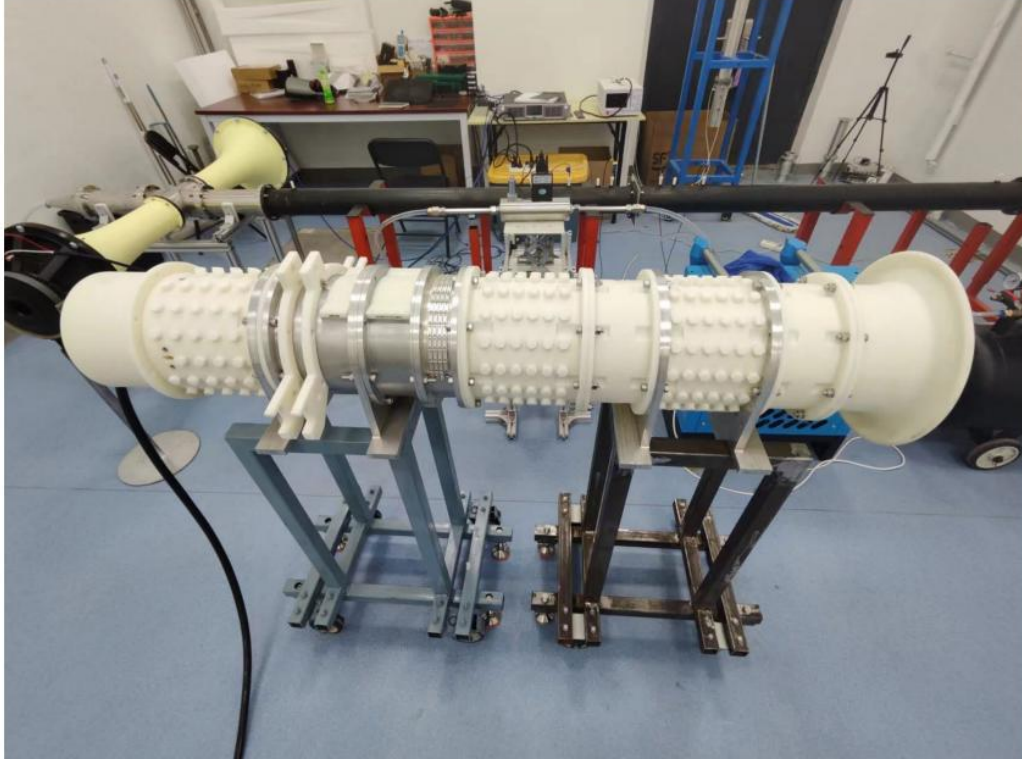


Figure 4.5. Fan aerodynamic-noise test rig

$L4 = 0.01$.

Reflection coefficients obtained using radial modal decomposition are compared with experimental results below.

4.3.2 Circumferential Decomposition

In the preceding section, points were placed along three cut lines at different radial positions, varying the radius r . Placing points at different circumferential positions on the wall instead varies the azimuthal angle θ . During solution of the modal-decomposition matrix system, the former arrangement is more resistant to disturbances than the latter, as discussed in detail in Section 4.3.4 on error bands. However, radial placement is difficult experimentally and disturbs the aerodynamics, so points are placed at different angles on the wall wherever possible.

The experiment uses 32 uniformly arranged measurement points distributed over four rings, as shown below.

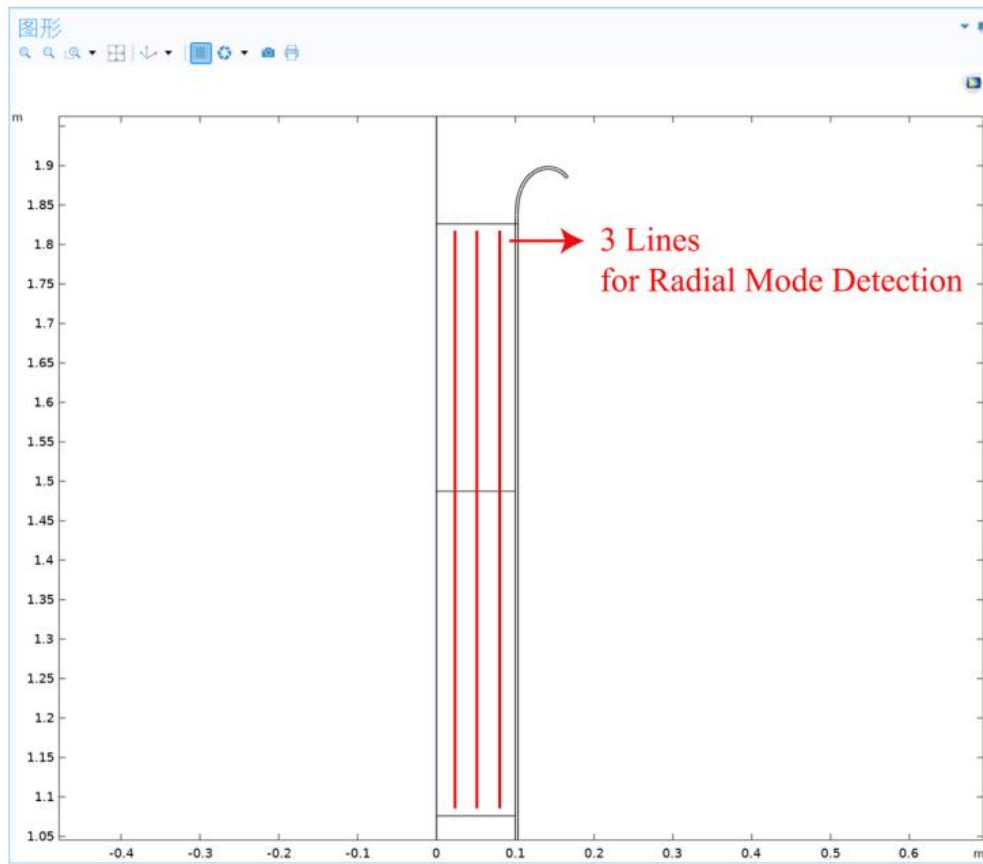


Figure 4.6. Data-extraction positions for radial modal decomposition

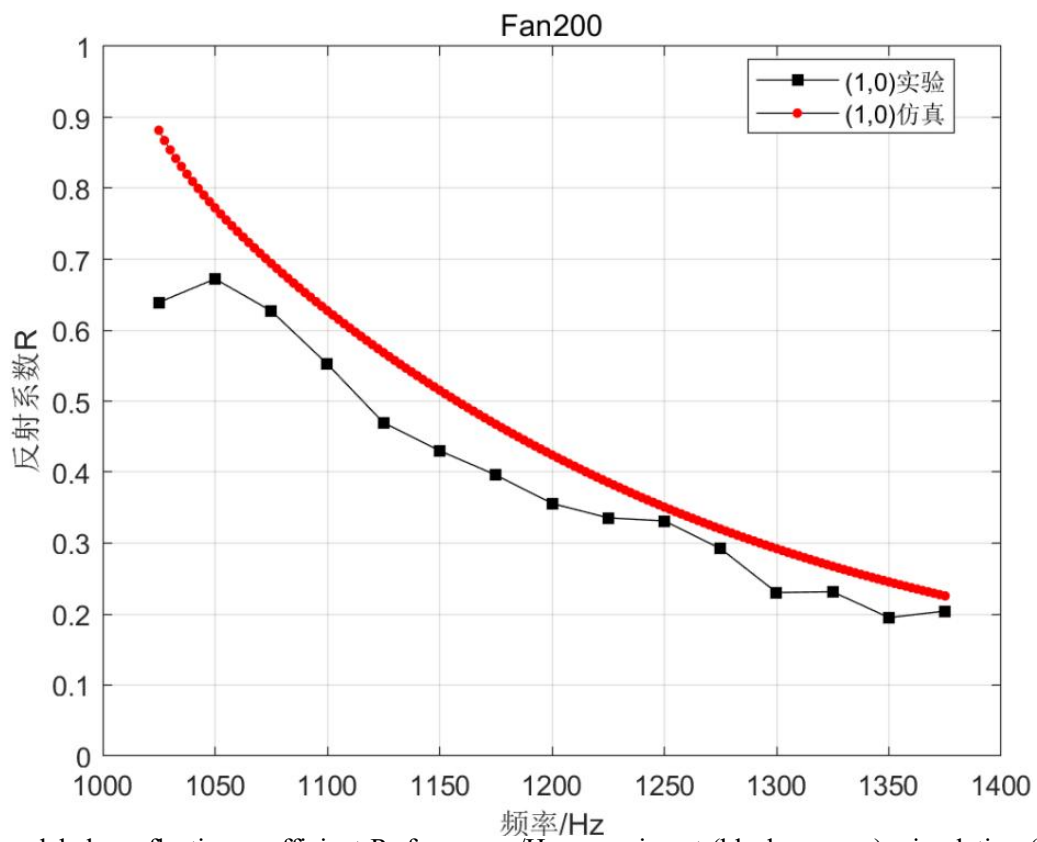


Image labels: reflection coefficient R; frequency/Hz; experiment (black squares); simulation (red circles), both for the (1,0) mode.

Figure 4.7. Reflection coefficients obtained using radial modal decomposition

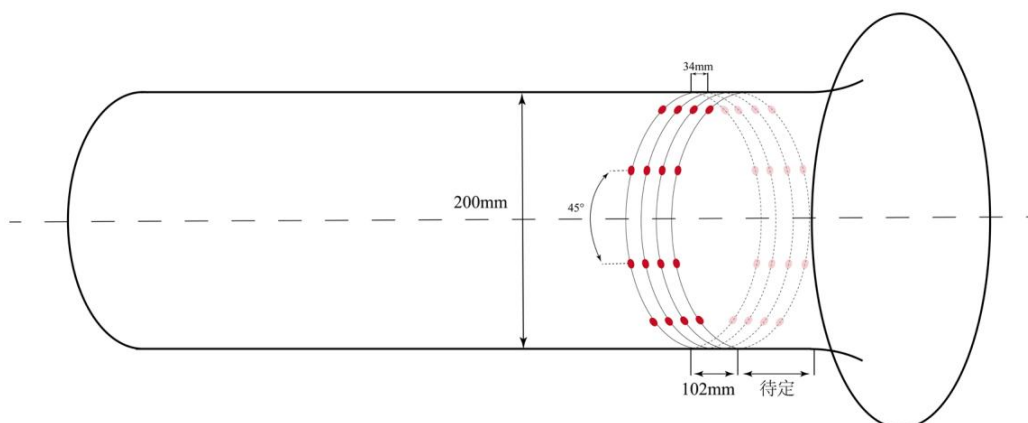


Image label at the downstream spacing: specified distance.

Figure 4.8. Circumferential measurement-point arrangement in the experiment

Corresponding points are selected in the simulation to extract the reflection coefficient:

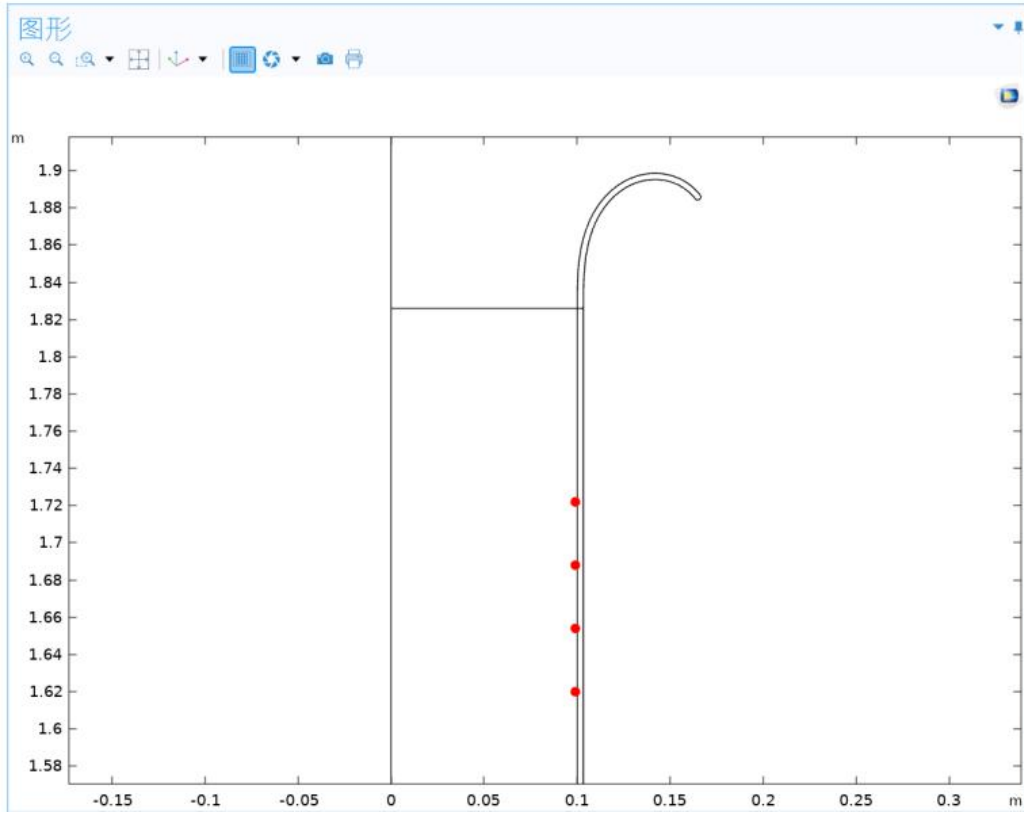


Figure 4.9. Measurement-point arrangement for circumferential modal decomposition

Sound pressure data are extracted at four points along the wall with a spacing of 34 mm. The pressure data $p(x, r)$ extracted from the two-dimensional axisymmetric model do not contain the circumferential phase information, namely the factor $e^{im\theta}$ in Equation (2.13). The exported $p(x, r)$ data must therefore be multiplied by $e^{im\theta}$, with $\theta = (0 : \frac{1}{4}\pi : 2\pi)$, to obtain $p(x, r, \theta)$.

The resulting data correspond point by point to the experimental data: four rings with eight points per ring, giving 32 locations in total. Circumferential modal decomposition is then used to extract the reflection coefficient, giving the results below.

The reflection coefficients obtained using circumferential decomposition are almost identical to those obtained using radial decomposition in the previous section. This is discussed in detail in the next section.

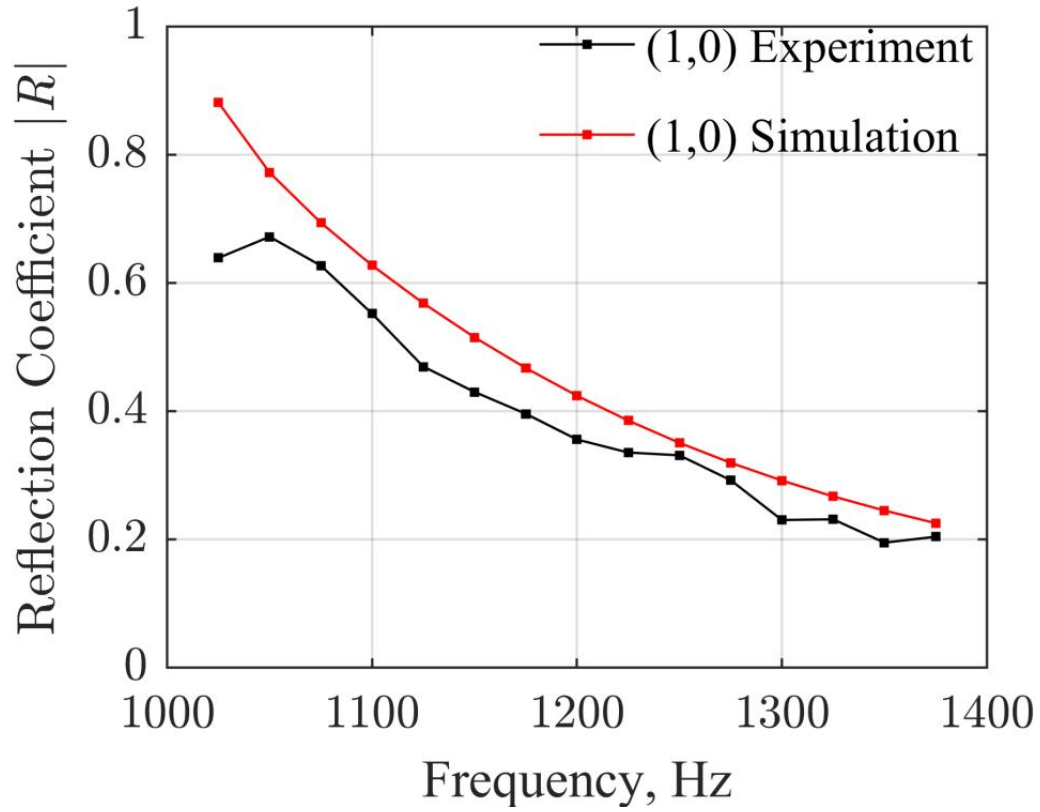


Figure 4.10. Reflection coefficients obtained using circumferential modal decomposition

4.3.3 Robustness

The preceding two sections investigated reflection-coefficient extraction using measurement points at different radial and circumferential positions. This raises the question of how the measurement-point arrangement affects extraction. Taking the reflection coefficients obtained from the four-ring, 32-point circumferential arrangement (Figure 4.8) as a baseline, this section investigates different radial arrangements while maintaining a total of 32 measurement points. The selected arrangements are listed below.

Table 4.4. Measurement-point arrangements

C4-8	R2-16	R4-8	R8-4
Circumferential decomposition	Radial decomposition	Radial decomposition	Radial decomposition
4 rings	2 columns	4 columns	8 columns
8 points per ring	16 points per column	8 points per column	4 points per column

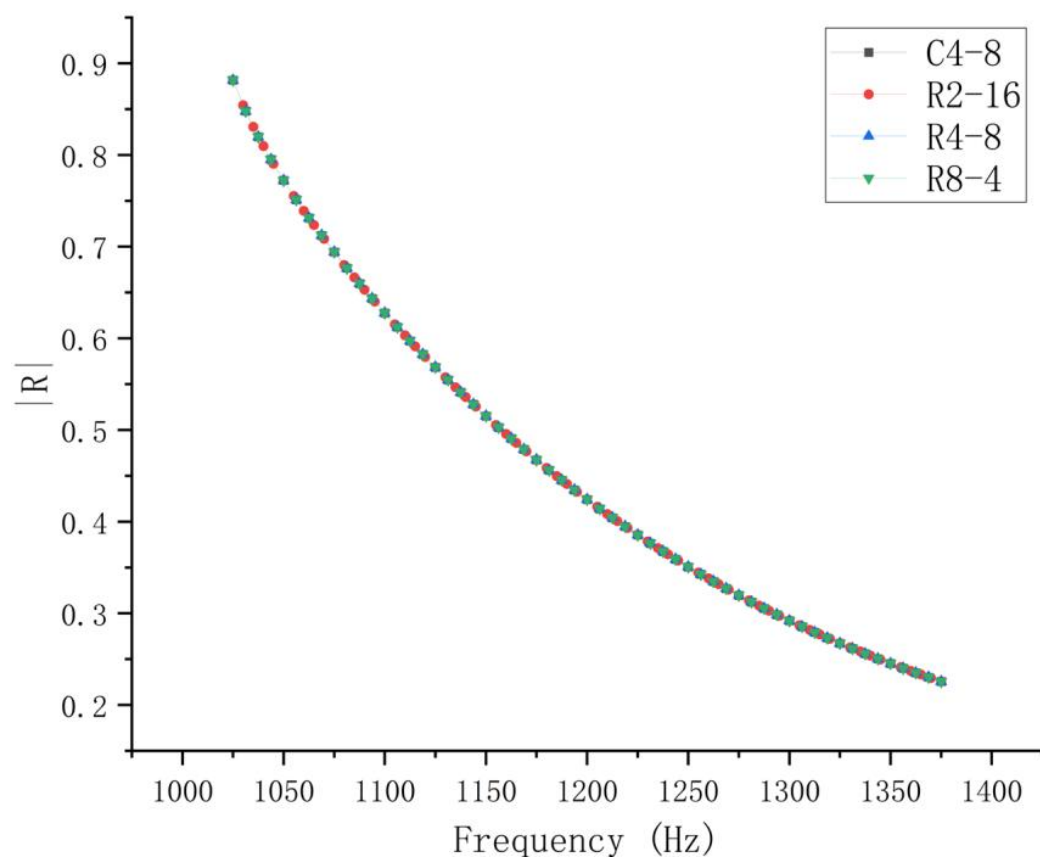


Figure 4.11. Reflection coefficients extracted using different measurement-point arrangements

As shown above, different arrangements have almost no effect on the extracted reflection coefficient when the total number of measurement points is 32. In experiments, however, measurements in the transition section, the straight section, or different positions within the straight section give different decomposition results because of flow disturbances. The present linearized potential-flow calculations do not account for the flow boundary layer. Consequently, extracting and decomposing data at different positions in the straight section of the computational model gives consistent results.

4.3.4 Error Bands

The previous section examined the robustness of reflection-coefficient extraction from sound pressures obtained using the same number of measurement points at different positions and decomposed circumferentially or radially. The reflection coefficients already agree reasonably closely with the experimental values. This section introduces error bands into the simulation results to improve their comparison with experimental data.

In COMSOL simulations, the pressure at each point can be calculated relatively accurately by the finite-element method. In experiments, however, microphone hardware introduces errors of ± 0.5 dB or ± 1 dB into the measured pressure at a point p_i . Sound pressure level errors must therefore be introduced into the simulation data. Each case is perturbed 100, 1000, 10000, or more times, and a 95.45% confidence interval is taken to assess whether discrepancies between experiment and calculation can be attributed to measurement error.

The following discussion examines how sound pressure level and phase-angle errors affect the error bands and how radial and circumferential decomposition methods influence their widths.

4.3.4.1 Introducing Error Terms

The sound pressure level in the COMSOL Acoustics Module is calculated as

$$SPL = 20 \log_{10} \frac{\frac{|p_{rz}|}{\sqrt{2}}}{2 \times 10^{-5}} dB \quad (4.16)$$

Here, p_{rz} is the exported complex sound pressure at coordinates (r, z) and contains phase information. To simulate experimental errors of ± 0.5 dB or ± 1 dB, these errors must first be converted into sound pressure errors, after which the corresponding reflection-coefficient error bands are calculated. The procedure is described below.

For an individual point, the SPL after a perturbation SPL_{Δ} is

$$SPL_{disturb} = 20 \log_{10} \left[\frac{\frac{|p_{rz}|}{\sqrt{2}}}{2 \times 10^{-5}} \times \left(\frac{p_e}{2 \times 10^{-5}} \right)^{random(-1, +1)} \right] \quad (4.17)$$

Where $p_e = 2 \times 10^{\frac{SPL_{\Delta}}{20} - 5} Pa$

The perturbed sound pressure magnitude is then

$$|p_{rz, disturb}| = |p_{rz}| \times 10^{\frac{random(-1, +1) SPL_{\Delta}}{20}} \quad (4.18)$$

The phase information is obtained from the phase angle of the COMSOL-exported pressure p_{rz} and is kept unchanged.

The perturbed complex sound pressure is finally obtained as

$$p_{rz, disturb} = |p_{rz, disturb}| \times \cos(\varphi) + |p_{rz, disturb}| \times \sin(\varphi)i \quad (4.19)$$

After introducing an amplitude perturbation, phase-angle errors must also be introduced when using circumferential modal decomposition. For pressures at different circumferential positions, phase perturbations are added as follows [21]:

$$p_{rz, disturb} = 10^{\frac{random(-1, +1) \cdot SPL_{\Delta}}{20}} |p_{rz}| e^{\{im[\theta_{rz} + random(-1, +1) \cdot \delta_{\theta}]\}} \quad (4.20)$$

Here, θ_{rz} is the phase angle at the point, m is the mode order, and δ_{θ} is the phase-angle perturbation. This can be understood as first adding an amplitude perturbation to the pressure at a point and then adding a phase perturbation.

4.3.4.2 Error Bands for Circumferential Modal Decomposition

The error-band procedure above is applied to C4-8: circumferential decomposition using four rings with eight points each. Perturbations of ± 1 dB and $\pm 1^\circ$ are prescribed, with 10^2 , 10^3 , and 10^4 perturbation realizations, respectively. The results are shown below.

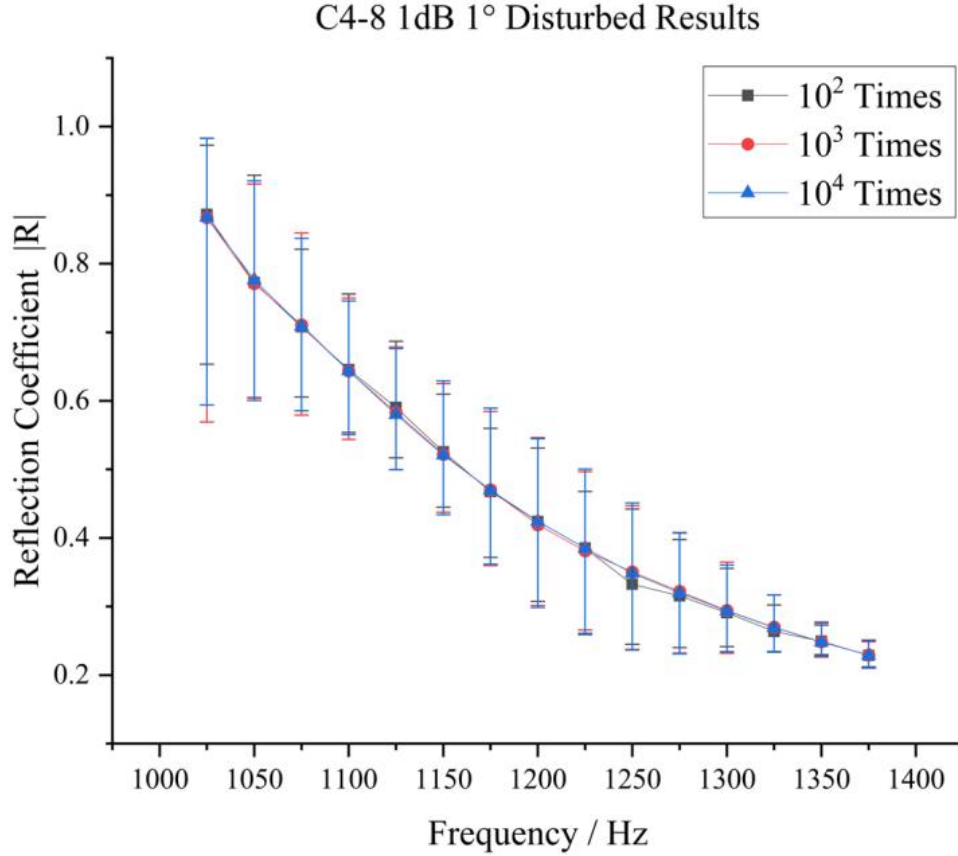
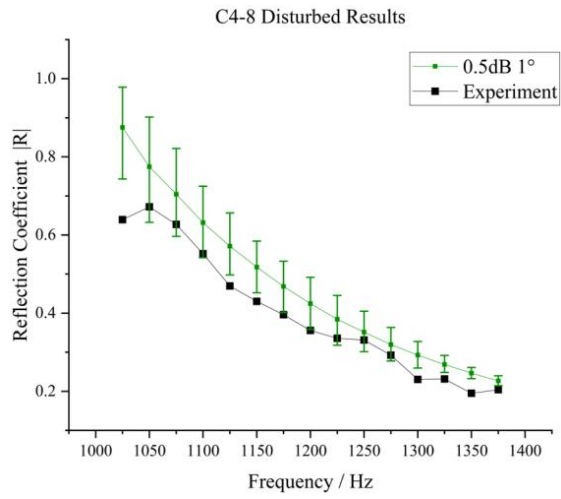


Figure 4.12. Error bands for different numbers of perturbation realizations

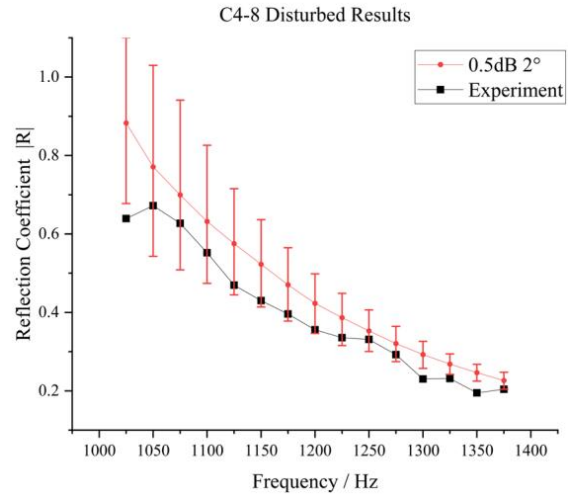
At the initial point, 1025 Hz, the error band is relatively wide because this frequency is close to cutoff. Thereafter, the error bands for different numbers of realizations are very similar, and increasing the number has little effect on their extent. Therefore, 1000 realizations are used in the subsequent calculations.

Table 4.5. Numbers of experimental data points covered by the error bands

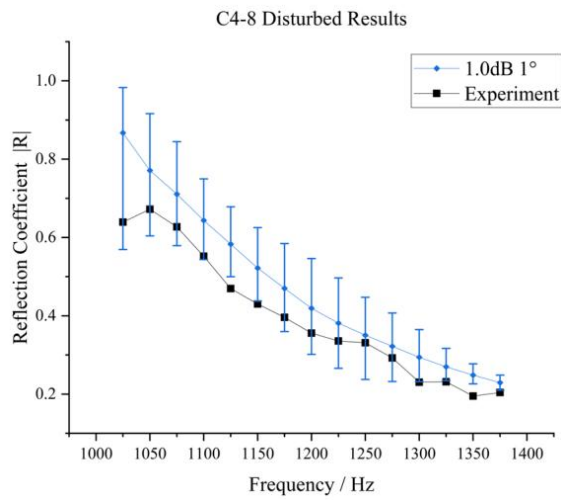
0.5dB +1°	0.5dB +2°	1.0dB +1°	1.0dB +2°
8 points	11 points	12 points	14 points



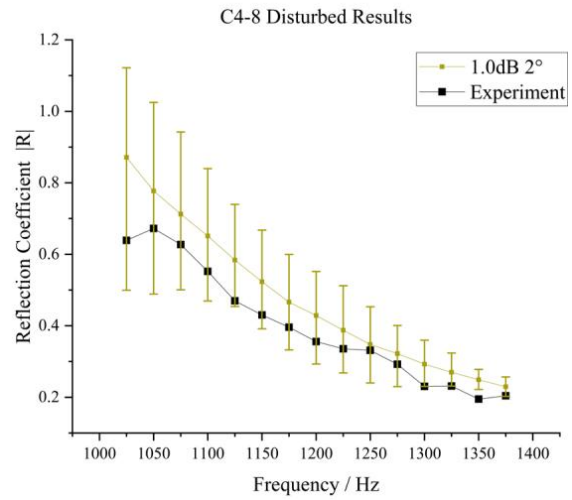
(a) 0.5 dB and 1°



(b) 0.5 dB and 2°



(c) 1.0 dB and 1°



(d) 1.0 dB and 2°

Figure 4.13. Error bands for different perturbation combinations

4.3.4.3 Errors in Radial Modal Decomposition

Section 4.3.3 discussed the robustness of reflection-coefficient extraction using different arrangements of 32 measurement points. Similarly, this section examines how the arrangement affects the extent of the error band.

Based on the experimental-data coverage in the preceding section, the perturbation combination $1.0 \text{ dB} + 2^\circ$, which covers the largest number of points, is selected. Error bands for different measurement-point arrangements are shown below.

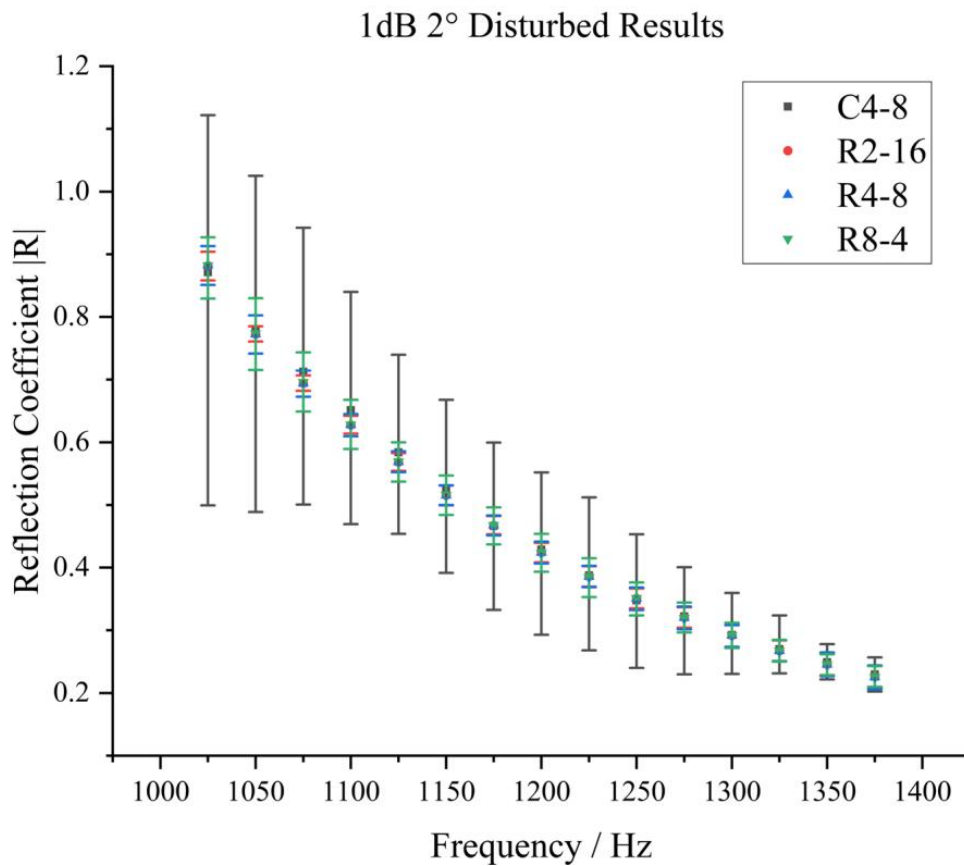


Figure 4.14. Reflection coefficients extracted using different measurement-point arrangements

For the same number of measurement points, the circumferential arrangement C4-8 gives a substantially wider error band after circumferential modal decomposition than the radial arrangements R2-16, R4-8, and R8-4 after radial modal decomposition. Differences among the radial arrangements are small. Their resistance to perturbations follows $R2-16 > R4-8 > R8-4 > C4-8$.

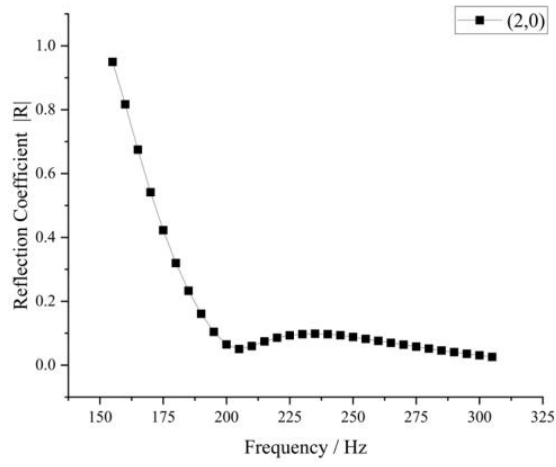
With the same number of measurement points, circumferential modal decomposition is less resistant to disturbances affecting the error bands than radial modal decomposition. In addition, the error bands obtained by both methods tend to narrow as frequency increases.

4.4 Extended Calculations

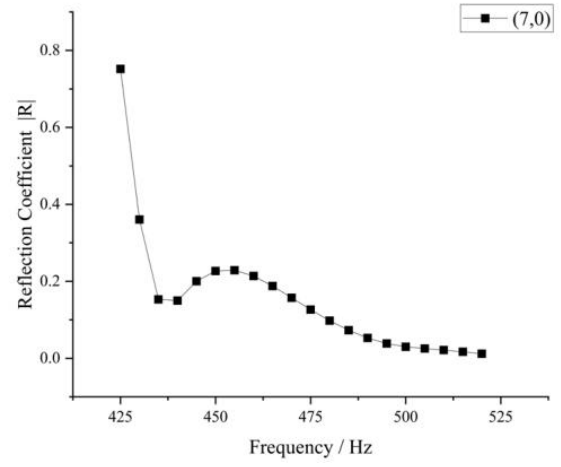
As noted in Chapter 1, sources in actual engines are often concentrated near the cut-off frequency. Sound pressure level measurements in this region may contain errors because of the large reflection coefficient. Furthermore, in engine nacelle noise reduction design, Zorumski's mode-matching method [27][28] can guide the design of multisection liners and calculate finite-length ducts, provided that the modal reflection coefficients are known. The bellmouth reflection-coefficient method in Section 4.3.1 is therefore applied to a high-bypass-ratio engine in an engineering configuration. The reflection coefficients for different modes are shown below.

4.5 Summary

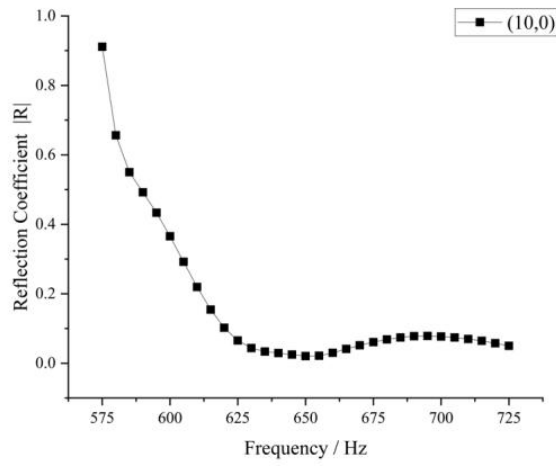
This chapter modeled and simulated a bellmouth duct using the COMSOL Linearized Potential Flow, Frequency Domain interface. Reflection coefficients were extracted using radial and circumferential modal decomposition. With the same number of measurement points, the radial and circumferential results were found to be robust to different measurement positions. Sound pressure level and phase-angle errors were then introduced into the simulation data. For particular perturbation combinations, the resulting error bands covered the experimental results reasonably well. The resistance of different measurement-point arrangements to perturbations was also investigated. However, the error bands tended to narrow as frequency increased. The next chapter further examines this observation.



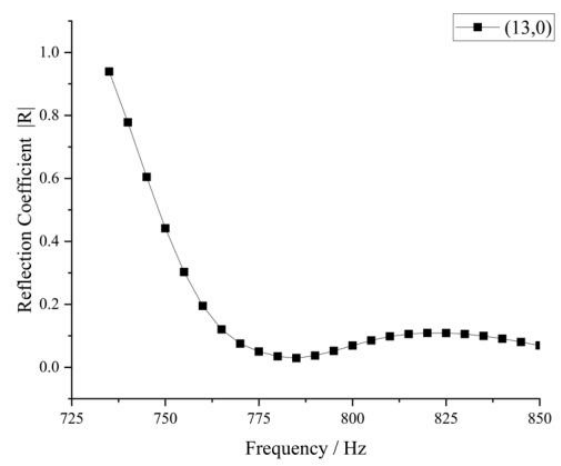
(a) (2,0) mode



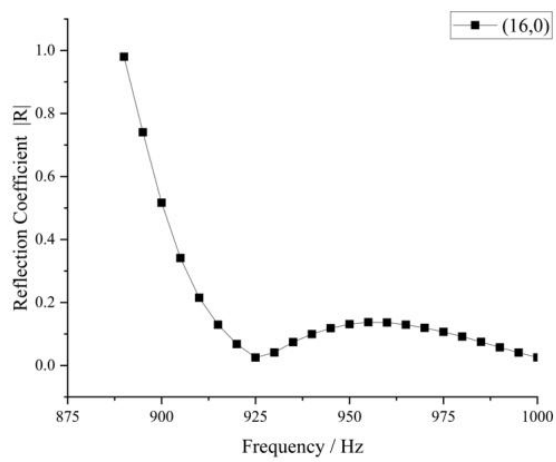
(b) (7,0) mode



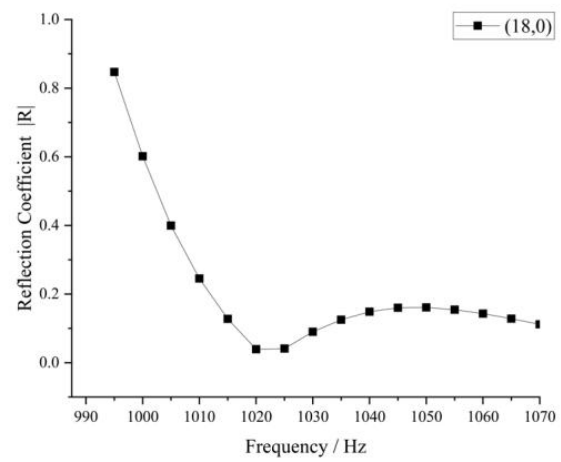
(c) (10,0) mode



(d) (13,0) mode



(e) (16,0) mode



(f) (18,0) mode

Figure 4.15. Reflection coefficients for different modes

5. Sound Field Reconstruction

In the error-band calculations of the previous chapter, the bands tended to narrow as frequency increased. This chapter considers sound field reconstruction as a way to explain this observation.

The incident sound pressure amplitude calculated by modal decomposition for C4-8, P_{mn}^+ in Equation (2.23), is compared with the incident amplitude prescribed in COMSOL, P_{in} in Section 4.2.1. The results are shown below.

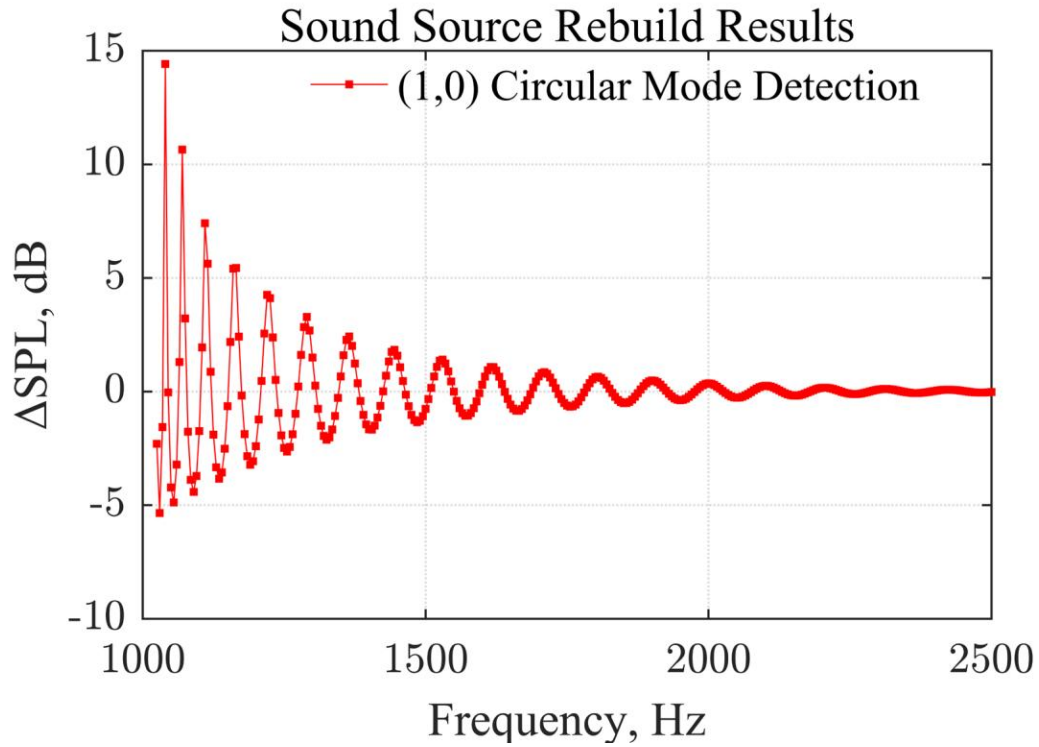


Figure 5.1. Sound field reconstruction results

At low frequencies, the calculated incident sound pressure amplitude differs substantially from the prescribed value. This discrepancy gradually decreases as frequency increases. The following sections analyze the observation from three perspectives.

5.1 Effect of Axial Spacing

As noted in Section 2.4.2, when measurement points are arranged axially, their spacing must satisfy the following condition to obtain accurate duct-mode decomposition results:

$$0.1\pi < \frac{ks}{1 - M^2} < 0.8\pi$$

Here, k is the axial wavenumber, s is the axial microphone spacing, and M is the Mach number. Applying this criterion gives the plots below.

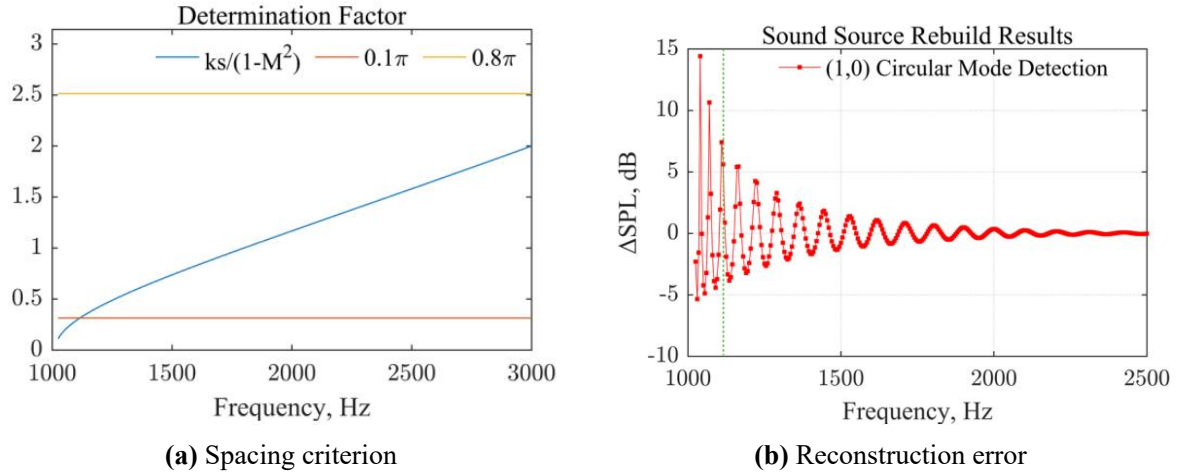


Figure 5.2. Effect of axial spacing

The horizontal coordinate of the intersection of the two lines at the lower left of panel (b) is 1115 Hz. Considered together with panel (b), indicated by the green dashed line in the right-hand plot, the oscillation amplitude of the sound pressure level deviation decreases above 1115 Hz, with a maximum deviation of 5.44 dB. This indicates that the interaction between measurement-point spacing and frequency affects the sound field reconstruction results.

5.2 Effect of Measurement-Point Distribution

When extracting sound-field data inside the duct, only four axial measurement points are used, with a spacing of 34 mm. At low frequencies, these four points provide limited information over a complete wavelength. Starting from this observation, this section examines the axial standing-wave curves together with the measurement-point positions.

Standing-wave profiles inside the duct are plotted in panels (a) and (b) below for the frequencies in Figure 5.1 with the largest positive and negative deviations, 1040 Hz and 1030 Hz, respectively, and a frequency with a deviation smaller than 1 dB, 1700 Hz. Profiles at

frequencies with the largest positive and negative sound pressure level deviations are then plotted in panels (c) and (d), respectively. Small circles indicate microphone positions.

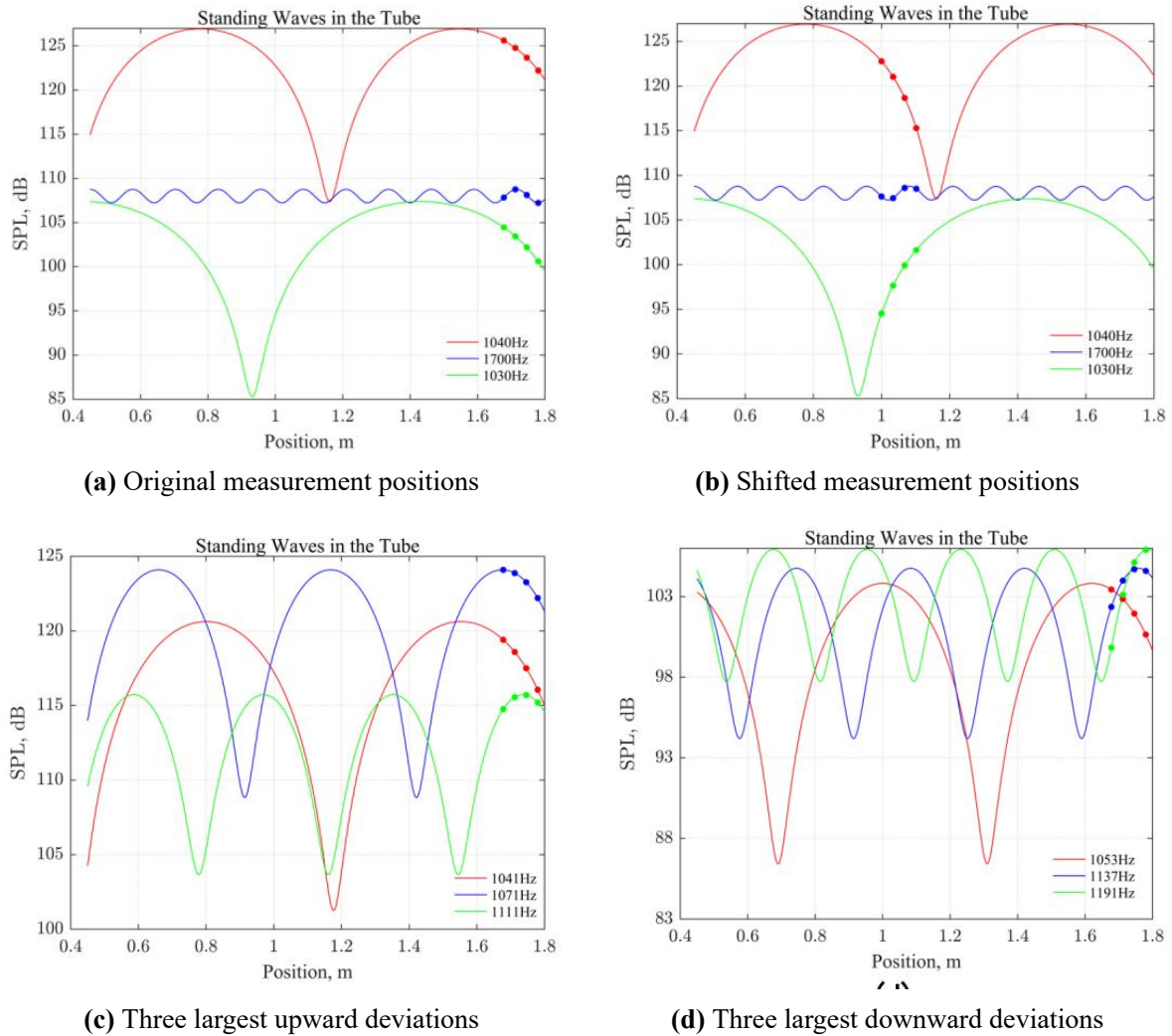


Figure 5.3. Effect of measurement-point distribution

Three frequencies with relatively large positive sound pressure level deviations

Three frequencies with relatively large negative sound pressure level deviations

Panels (a) and (b) show that at higher frequencies the four axial measurement points can cover a wavelength, whereas at lower frequencies they cover only a small fraction of a wavelength. Panels (c) and (d) also show a clear difference in standing-wave amplitudes between cases with large positive and negative sound pressure level deviations. For a positive deviation, the maximum sound pressure level approaches 125 dB, whereas for a negative

deviation it reaches only 105 dB. This indicates a relationship between the deviation and the standing-wave sound pressure level amplitude. The next chapter discusses the effect of the reflection coefficient on these standing waves at different frequencies.

As shown in panel (b), the distribution of microphones relative to the standing wave can also be adjusted by changing their axial positions. The adjusted positions are listed below.

Table 5.1. Measurement-point positions

Configuration	C 34-1.78	C 34-1.212	C 136-1.78	C NU-1.78
Point spacing/mm	34	34	136	136, 68, 34
Coordinate of final point/m	1.78	1.212	1.78	1.78

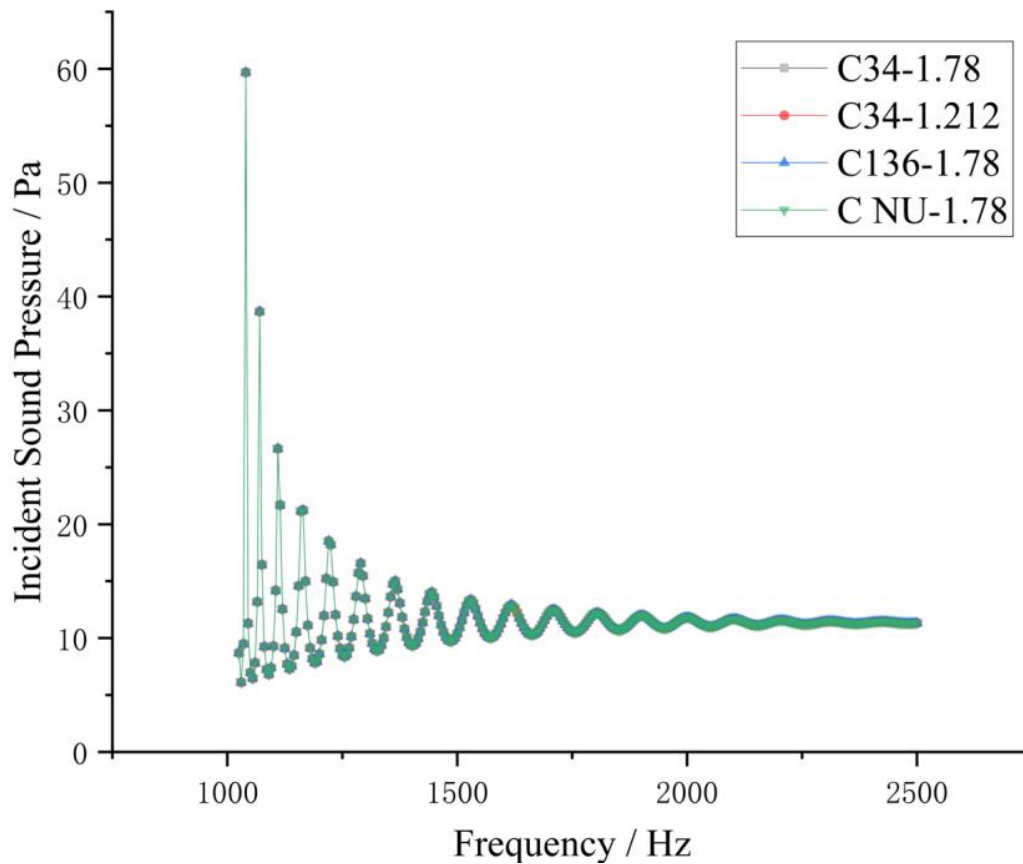


Figure 5.4. Sound field reconstruction for different measurement-point distributions

As shown above, circumferential modal decomposition of sound pressure data from the different measurement-point distributions gives identical incident amplitudes. The sound

field reconstruction deviation is therefore independent of the axial distribution of the measurement points.

5.3 Effect of the Reflection Coefficient

In the preceding section, the sound field reconstruction results consistently exhibited oscillations at low frequencies, with amplitudes that gradually decreased as frequency increased. This is consistent with the trend of the reflection coefficient. This section therefore considers the reconstruction results together with the reflection coefficient.

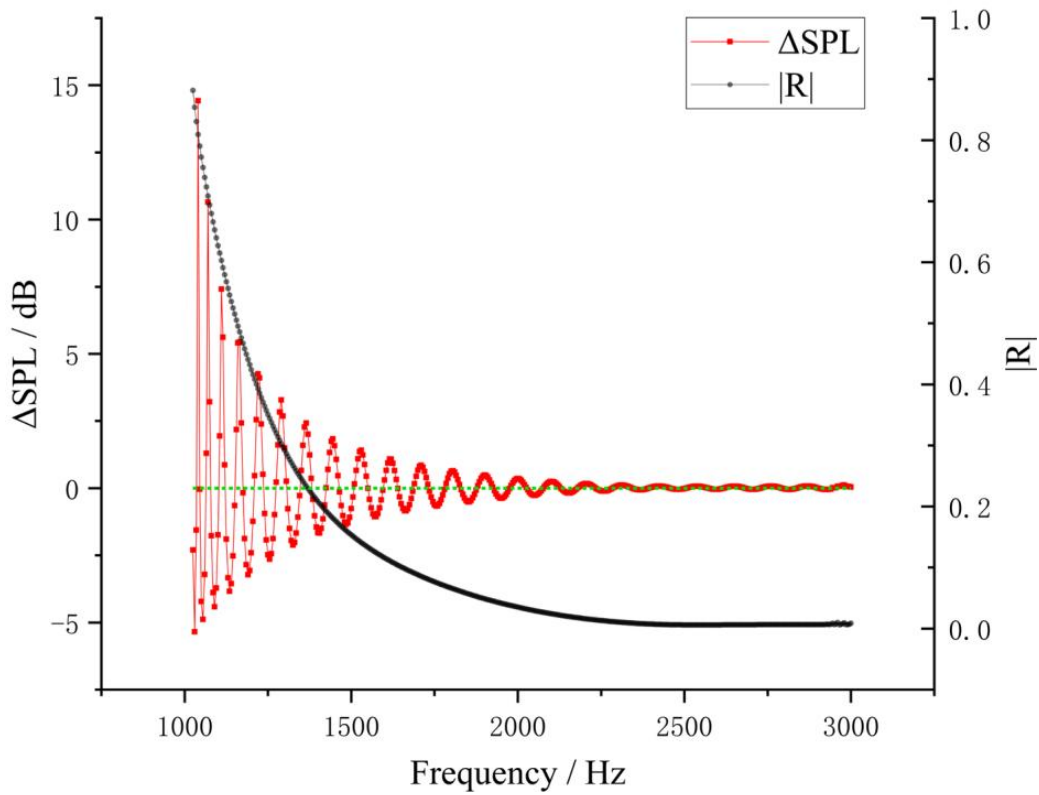


Figure 5.5. Sound field reconstruction results considered together with the reflection coefficient

The reconstruction deviation gradually decreases as the reflection coefficient decreases. Closer to the cutoff frequency, the reflection coefficient is larger and the corresponding reconstruction deviation is also larger. To illustrate this further, the duct outlet is changed to a straight opening. In this configuration, the reflection coefficient decreases more slowly with frequency than for the bellmouth, as shown below.

Correspondingly, as shown in Figures 9 and 10, the oscillations in the reconstruction

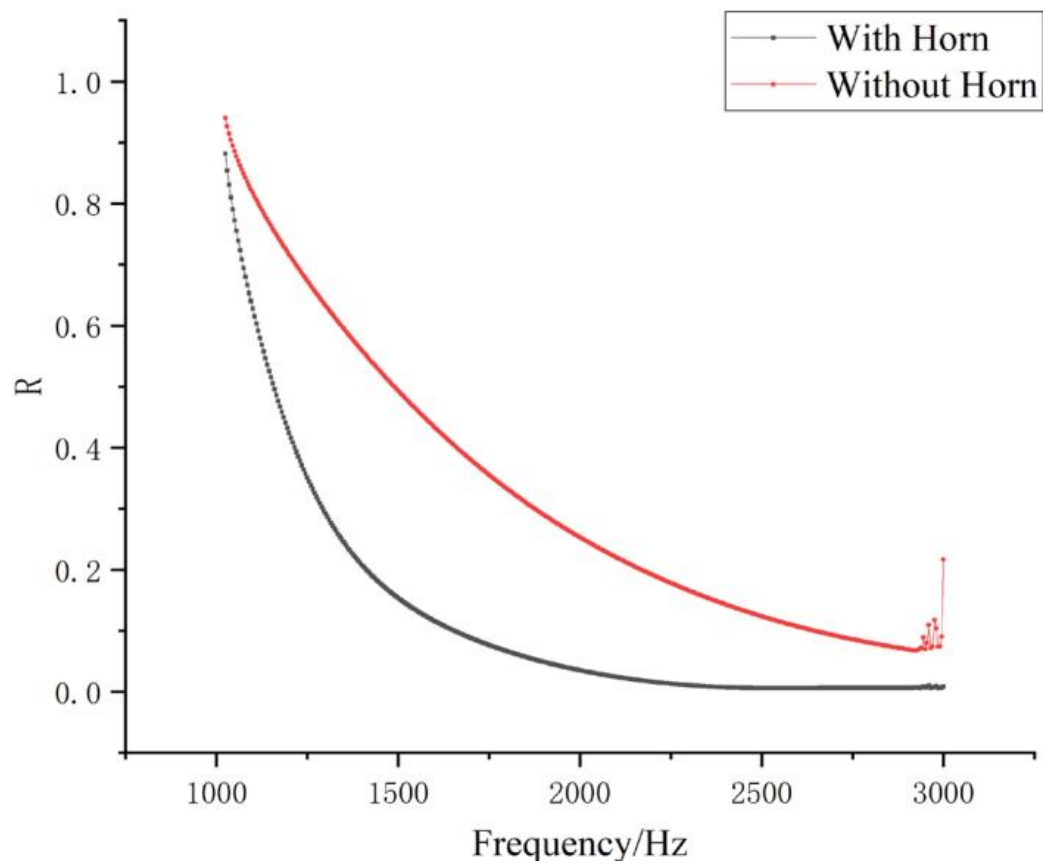


Figure 5.6. Reflection coefficients for ducts with and without a bellmouth

deviation decay more slowly with frequency for the straight-opening duct than for the bell-mouth duct, whose reflection coefficient decreases more rapidly.

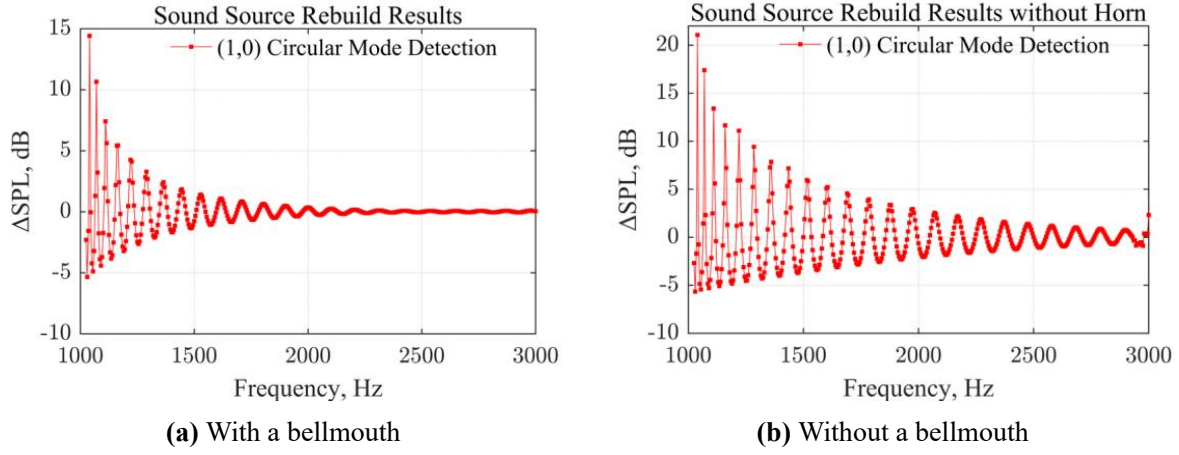


Figure 5.7. Sound field reconstruction for ducts with and without a bellmouth

5.4 Summary

This chapter preliminarily analyzed the large low-frequency oscillations in sound field reconstruction from three perspectives. Measurement-point spacing and changes in the reflection coefficient may contribute to these oscillations. Furthermore, all the preceding analyses use COMSOL simulation data. As described in Section 4.2.2, a PML boundary condition was applied at the duct inlet in the model. PML absorption can be less effective at low frequencies or in the presence of evanescent waves, which may also contribute to the oscillations. Their specific causes require further investigation.

6. Standing-Wave Results inside the Duct

The main objective of this chapter is to obtain axial standing-wave distributions inside the bellmouth duct from the simulation data of Chapter 5 and then measure standing waves in the actual duct experimentally.

6.1 Simulated Standing-Wave Results

Sound pressure levels are extracted at different distances from the opening inside the duct and plotted against frequency below.

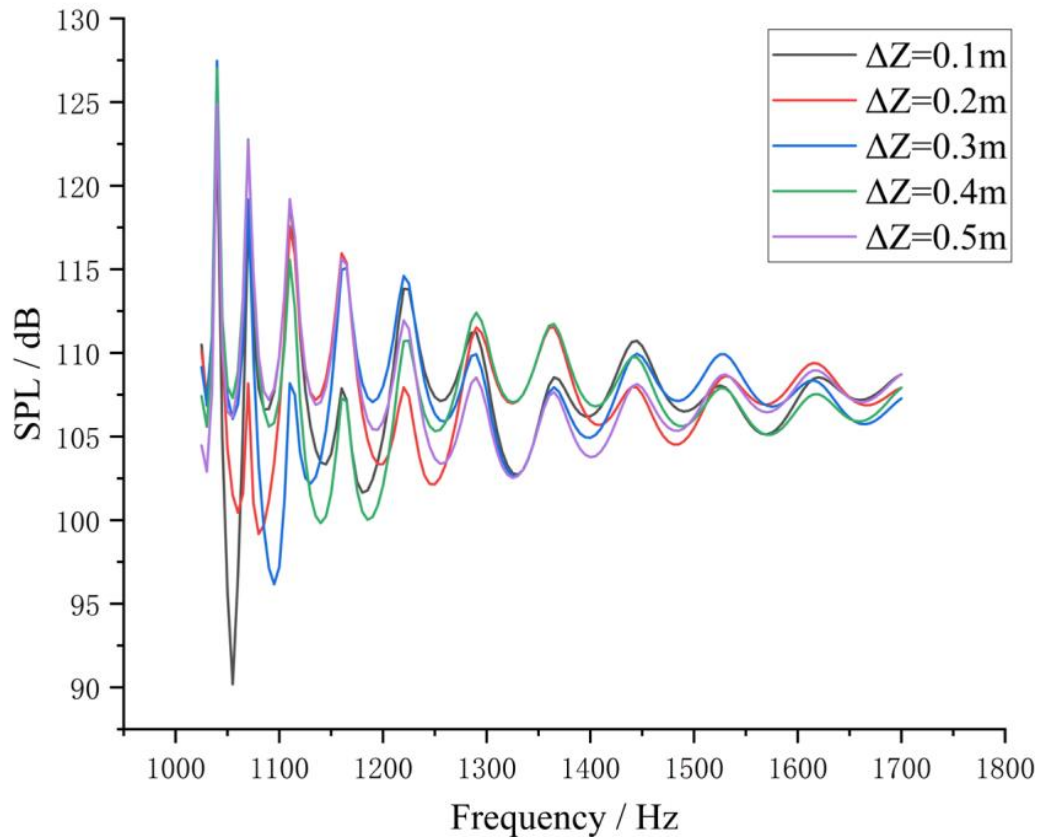


Figure 6.1. Standing-wave results inside the duct

As shown above, the spacing in frequency between adjacent peaks increases at higher frequencies because it is influenced by the axial wavenumber k_z . As frequency increases, k increases, while the radial wavenumber k_r remains constant for a given mode. The corresponding k_z therefore increases, and the peak spacing increases. The fluctuation amplitude is also influenced by the reflection coefficient: at higher frequencies the reflection coefficient

is smaller, so the fluctuations become smaller.

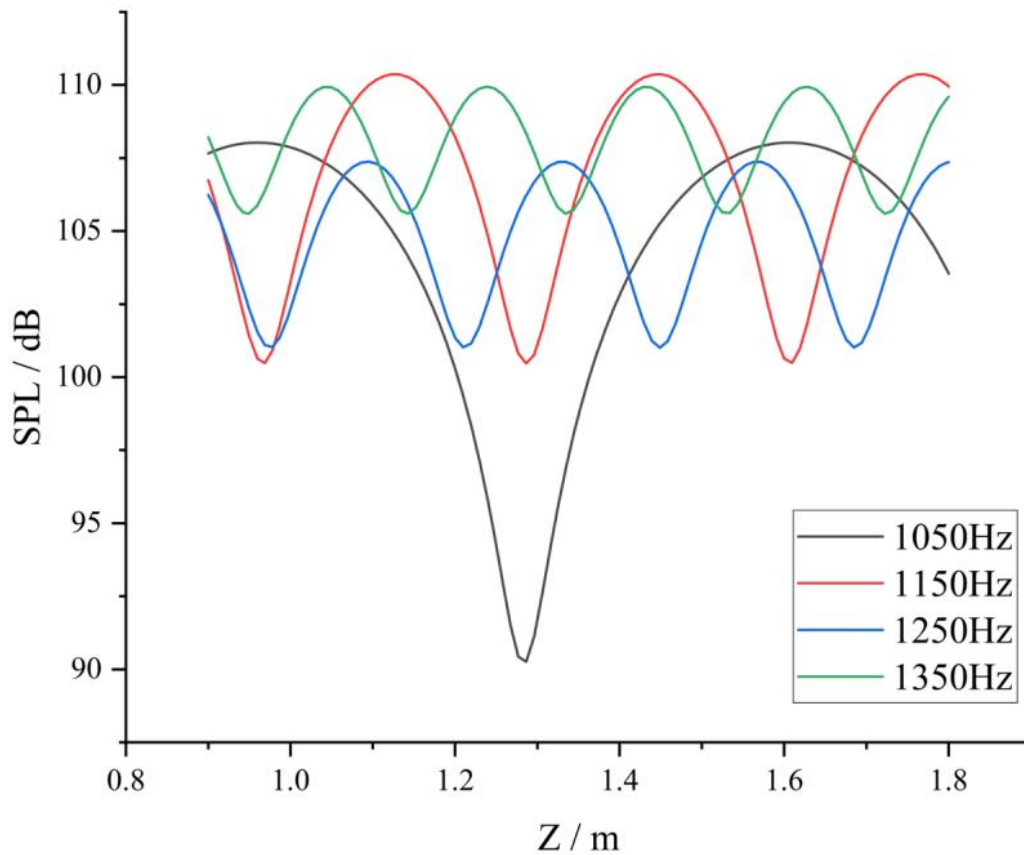


Figure 6.2. Standing-wave results inside the duct at selected frequencies

The sound pressure level at different positions for a fixed frequency is shown above, revealing clear standing waves. Here, 1) the wavelength is determined by frequency; 2) the mean level of the fluctuations is determined by source strength; and 3) the fluctuation amplitude is determined by the reflection coefficient. For the superposition of two waves, the upper limit corresponds to a doubling of sound pressure, or a 6 dB increase in SPL. If the waves cancel, as may occur with a relatively large reflection coefficient, the resulting pressure becomes very small and the SPL decreases sharply, producing the deep minimum shown above at 1050 Hz. The fluctuation range is also affected by the measurement position relative to the duct opening.

Standing-wave profiles are plotted below with the reflection coefficients and frequencies considered together.

Table 6.1. Reflection coefficients from 1050 to 1350 Hz

Frequency/Hz	Experimental reflection coefficient	Simulated reflection coefficient
1050	0.6709	0.772359
1100	0.5528	0.627526
1150	0.4302	0.515287
1175	0.3977	0.467412
1250	0.3286	0.350845
1300	0.2303	0.292128
1350	0.1951	0.245431

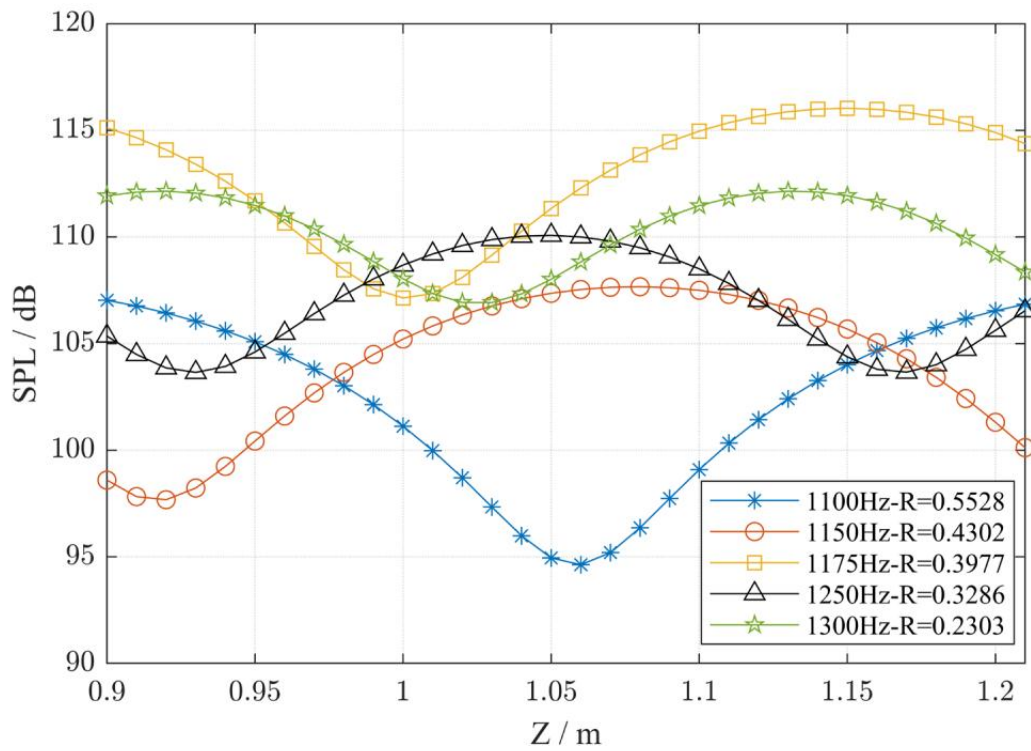


Figure 6.3. Standing-wave results at 10 measurement points with 10 mm spacing

6.2 Standing-Wave Measurement Experiment

The previous section obtained standing-wave profiles inside the duct from simulation data. This section extracts standing waves experimentally.

The test rig currently has 32 microphones with a diameter of 8 mm. Considering the size of their mounts, the minimum spacing should be no less than 20 mm. More than 10 microphones are required to resolve a waveform clearly. Furthermore, a larger reflection coefficient makes it harder to capture the sharp minima shown in Figure 6.2, requiring a denser microphone arrangement.

1) Test Frequencies

Because the experimental motor has different stability characteristics at different frequencies, the test band is limited to 200 Hz. Two or three frequencies suitable for observing standing waves are selected between 1100 and 1300 Hz, corresponding to motor speeds of 4400–5200 rpm for 15 blades. One or two waveforms should be measurable over the available length.

During the experiment, the waveform needs to be observed in real time. Fine adjustments to frequency can help capture a well-resolved waveform, particularly its minima. The desired waveform has a relatively large distance between peaks and troughs, with troughs that are not excessively sharp.

2) Measurement-Point Arrangement

The test rig has good circumferential symmetry, with sound pressure level differences around a ring within 1 dB. Two staggered axial rows of microphones can therefore be used to reduce effective spacing while facilitating installation.

The measurement-point arrangement is illustrated below.

Two axial rows of microphones are mounted on the casing. To facilitate installation, a spacing of 20 mm is assumed between microphones on the same axial line and 10 mm between adjacent microphones on different lines. This produces a combined sequence of measurement points with 10 mm axial spacing.

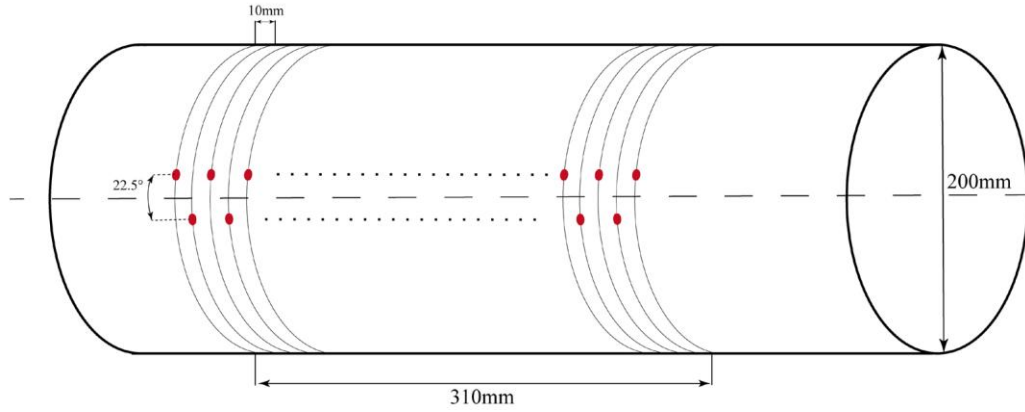


Figure 6.4. Schematic of measurement points for standing-wave measurements

The corresponding casing model is shown below.

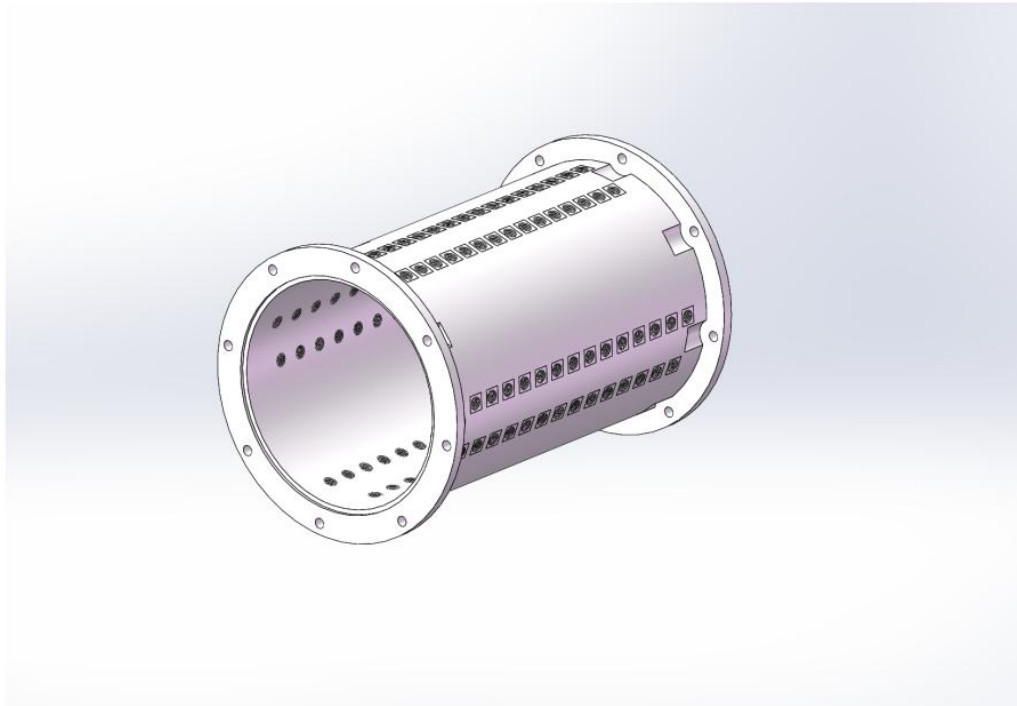


Figure 6.5. Casing model for standing-wave measurements

Eight rows of microphone mounting holes are provided, with each pair of adjacent rows forming a group. The within-row spacings for the four groups are 20 mm, 22 mm, 24 mm, and 26 mm. Each mounting hole is a GB-standard M12 threaded hole, with a flat seating surface to facilitate assembly.

GRAS_46BD 1/4-inch microphones are used in the experiment. Their measurement error is within ± 1 dB over 10 Hz–25 kHz. Matching microphone mounts are designed to

facilitate installation and keep the microphone faces flush with the duct wall.



Figure 6.6. GRAS_46BD 1/4-inch microphone

The experimental installation is shown below.

3) Experimental Results

The measured standing-wave profiles at different frequencies are shown below.

The results resemble those in Figures 6.1.2 and 6.1.3 and show clear standing waves. Here, 1) the standing-wave wavelength is determined by frequency; and 2) the fluctuation amplitude is determined by the reflection coefficient. As frequency increases, the reflection coefficient and fluctuation amplitude gradually decrease.

The axial sound pressures are collected along two axial measurement lines. Because these lines have different circumferential positions, their sound pressure levels differ slightly, producing the small fluctuations visible in the waveforms. These fluctuations remain within 1 dB.

6.3 Summary

This chapter studied standing waves inside the duct using simulations and experiments. Sound pressure levels at different positions for a fixed frequency showed clear standing waves. Here, 1) the wavelength is determined by frequency; 2) the mean level of the fluc-



Figure 6.7. Microphone installation and measurement-point arrangement

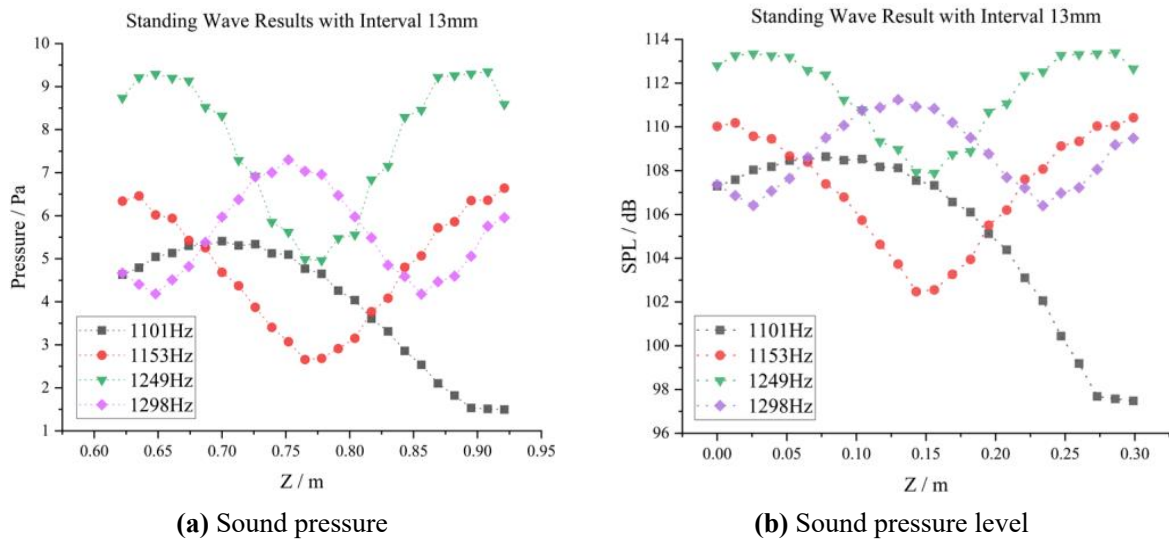


Figure 6.8. Experimental standing-wave measurement results

tuations is determined by source strength; and 3) the fluctuation amplitude is determined by the reflection coefficient. When two waves are superposed, the upper limit corresponds to a doubling of sound pressure, or a 6 dB increase in SPL. If the two waves cancel, as may occur with a relatively large reflection coefficient, the resulting pressure becomes very small and the SPL decreases sharply.

7. Conclusions and Outlook

This thesis used the Pressure Acoustics and Linearized Potential Flow interfaces of the commercial COMSOL finite-element solver to simulate ducts with different opening geometries: flanged ducts with different flange radii and a bellmouth duct. The Pressure Acoustics, Frequency Domain interface was used to model flanged ducts with different radii. Reflection coefficients calculated from the resulting sound pressure data using acoustic impedance integration agreed very well with published results. The two-point transfer-function method was also used to calculate reflection-coefficient curves at different positions, but because it processes data from only two points, its results differed substantially from experimental results. The Linearized Potential Flow, Frequency Domain interface was subsequently used to model the bellmouth duct, and reflection coefficients were extracted by radial and circumferential modal decomposition. With the same number of measurement points, radial and circumferential decomposition results were found to be robust to different measurement positions. Sound pressure level and phase-angle errors were then introduced into the simulation data, and the resulting error bands covered the experimental results reasonably well. However, the bands tended to narrow as frequency increased. A sound field reconstruction check was performed to investigate this observation, revealing large low-frequency oscillations whose amplitudes gradually decreased with increasing frequency. This may explain why the error bands are wider at low frequencies than at high frequencies. Preliminary analyses of measurement-point spacing, measurement-point distribution, and the reflection coefficient suggested that spacing and changes in the reflection coefficient may contribute to the oscillations. Poorer absorption by the inlet PML at low frequencies or in the presence of evanescent waves may also be a contributing factor.

The main conclusions are as follows:

- 1) Reflection coefficients obtained from COMSOL finite-element simulations agree well with published data, supporting the feasibility of the simulation method.
- 2) For the same number of measurement points, reflection coefficients obtained by ra-

dial and circumferential modal decomposition are robust. After error perturbations are introduced, however, radial decomposition is substantially more resistant to disturbances than circumferential decomposition. The reflection-coefficient error bands are also more sensitive to phase-angle errors than to sound pressure level errors.

3) After error bands are introduced, the simulation results encompass the experimental results. This supports both the accuracy achieved in the experiment and the feasibility of the simulation method.

4) Standing-wave profiles at different frequencies were obtained through simulations and experiments, and these profiles are affected by the reflection coefficient.

Future work can proceed in two directions: simulation studies and experimental measurements.

First, although the COMSOL Linearized Potential Flow interface can now simulate the bellmouth duct used in the experiment, this thesis has not yet provided a sufficiently satisfactory explanation for the low-frequency oscillations in the reconstructed sound field or the decrease in error-band width with increasing frequency. Future experiments on the corresponding duct could provide data for the same processing procedure. This would eliminate the influence of the simulated inlet PML boundary condition and test the correctness of the modal-decomposition code.

Second, further standing-wave measurements inside the duct are needed. The minimum installation distance between microphones makes it difficult to obtain a smooth standing-wave profile. This thesis used two staggered rows of microphones to acquire axial sound pressure levels, but this approach inevitably introduces level differences associated with their different circumferential positions. Future work could further reduce the circumferential separation while maintaining installation quality, thereby reducing circumferential errors. Sources capable of generating different modes could also be designed to measure modal reflection coefficients and standing-wave profiles inside the duct and compare them with simulation results.

References

- [1] J. I. Hileman. FAA research on aviation noise. In *32nd Annual Aviation Noise and Emissions Symposium*, 2018.
- [2] L. Bertsch, D. G. Simons, and M. Snellen. Aircraft Noise: The Major Sources, Modelling Capabilities, and Reduction Possibilities. IB 224-2015 A 110, German Aerospace Center (DLR), 2015.
- [3] D. L. Sutliff. A 20 year retrospective of the advanced noise control fan: contributions to turbofan noise research. In *AIAA Propulsion and Energy 2019 Forum*, 2019. DOI: 10.2514/6.2019-3824.
- [4] P. D. Joppa. An acoustic mode measurement technique. In *9th AIAA/NASA Aeroacoustics Conference*, Williamsburg, VA, 1984. DOI: 10.2514/6.1984-2337.
- [5] U. W. Ganz, P. D. Joppa, T. J. Patten, and D. F. Scharpf. Boeing 18-Inch Fan Rig Broadband Noise Test. NASA/CR-1998-208704, NASA, 1998.
- [6] E. R. Rademaker and P. Sijtsma. Mode detection with an optimised array in a model turbofan engine intake at varying shaft speeds. In *7th AIAA/CEAS Aeroacoustics Conference*, 2001. DOI: 10.2514/6.2001-2181.
- [7] H. Levine and J. Schwinger. On the radiation of sound from an unflanged circular pipe. *Physical Review*, 73(4):383–406, 1948. DOI: 10.1103/PhysRev.73.383.
- [8] Y. Nomura, I. Yamamura, and S. Inawashiro. On the acoustic radiation from a flanged circular pipe. *Journal of the Physical Society of Japan*, 15(3):510–517, 1960. DOI: 10.1143/JPSJ.15.510.
- [9] R. Caussé, J. Kergomard, and X. Lurton. Input impedance of brass musical instruments: comparison between experiment and numerical models. *The Journal of the Acoustical Society of America*, 75(1):241–254, 1984. DOI: 10.1121/1.390402.
- [10] F. Silva, P. Guillemain, J. Kergomard, B. Mallaroni, and A. N. Norris. Approximation formulae for the acoustic radiation impedance of a cylindrical pipe. *Journal of Sound and Vibration*, 322:255–263, 2009. DOI: 10.1016/j.jsv.2008.11.008.
- [11] A. N. Norris and I. C. Sheng. Acoustic radiation from a circular pipe with an infinite flange. *Journal of Sound and Vibration*, 135(1):85–93, 1989. DOI: 10.1016/0022-460X(89)90756-6.
- [12] Y. Ando. On the sound radiation from semi-infinite circular pipe of certain wall thickness. *Acustica*, 22:219–225, 1969.
- [13] J.-P. Dalmont, C. J. Nederveen, and N. Joly. Radiation impedance of tubes with different flanges: numerical and experimental investigations. *Journal of Sound and Vibration*, 244(3):505–534, 2001. DOI: 10.1006/jsvi.2000.3487.
- [14] A. R. Silva, P. H. Mareze, and A. Lenzi. Approximate expressions for the reflection coefficient of ducts terminated by circular flanges. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 34(2):219–224, 2012. DOI: 10.1590/S1678-58782012000200014.
- [15] J. Premo and P. Joppa. Fan noise source diagnostic test: wall measured circumferential array mode results. In *8th AIAA/CEAS Aeroacoustics Conference and Exhibit*, 2002. DOI: 10.2514/6.2002-2429.

- [16] J. M. Ville and R. J. Silcox. Experimental Investigation of the Radiation of Sound from an Unflanged Duct and a Bellmouth, Including the Flow Effect. NASA-TP-1697, NASA, 1980.
- [17] M. L. Munjal. *Acoustics of Ducts and Mufflers with Application to Exhaust and Ventilation System Design*. John Wiley & Sons, 1987.
- [18] J. M. Tyler and T. G. Sofrin. Axial flow compressor noise studies. *SAE Transactions*, 70:309–332, 1962. DOI: 10.4271/620532.
- [19] U. Tapken and L. Enghardt. Optimisation of sensor arrays for radial mode analysis in flow ducts. In *12th AIAA/CEAS Aeroacoustics Conference*, 2006. DOI: 10.2514/6.2006-2638.
- [20] K. Shin and J. K. Hammond. *Fundamentals of Signal Processing for Sound and Vibration Engineers*. John Wiley & Sons, 2008. ISBN: 9780470511886.
- [21] L. Wang. 风扇管道声模态识别的实验及数值模拟研究, Northwestern Polytechnical University, 2019. In Chinese. Translated title: Experimental and Numerical Study of Acoustic Mode Identification in Fan Ducts.
- [22] M. Åbom. Modal decomposition in ducts based on transfer function measurements between microphone pairs. *Journal of Sound and Vibration*, 135(1):95–114, 1989. DOI: 10.1016/0022-460X(89)90757-8.
- [23] J. Su, D. Yang, and A. S. Morgans. Low-frequency acoustic radiation from a flanged circular pipe at an inclined angle. *The Journal of the Acoustical Society of America*, 151(2):1142–1157, 2022. DOI: 10.1121/10.0009586.
- [24] T. J. R. Hughes. *The Finite Element Method: Linear Static and Dynamic Finite Element Analysis*. Prentice-Hall, 1987.
- [25] X. Qiu, B. Xin, L. Wu, Y. Meng, and X. Jing. Investigation of straightforward impedance education method on single-degree-of-freedom acoustic liners. *Chinese Journal of Aeronautics*, 31(12):2221–2233, 2018. DOI: 10.1016/j.cja.2018.08.014.
- [26] L. Chen, L. Du, X. Wang, X. Jing, and X. Sun. A three-dimensional straightforward method for liner impedance education in uniform grazing flow. *Journal of Sound and Vibration*, 468:115119, 2020. DOI: 10.1016/j.jsv.2019.115119.
- [27] W. E. Zorumski. Acoustic Theory of Axisymmetric Multisectioned Ducts. NASA-TR-R-419, NASA Langley Research Center, 1974.
- [28] W. E. Zorumski. Generalized radiation impedances and reflection coefficients of circular and annular ducts. *The Journal of the Acoustical Society of America*, 54(6):1667–1673, 1973. DOI: 10.1121/1.1914466.